



THE UNITED REPUBLIC OF TANZANIA

GOVERNMENT ELECTRONIC PAYMENT GATEWAY APPLICATION PROGRAMMING INTERFACE FOR SERVICE PROVIDER (GePGSPAPIv5.0) TECHNICAL SPECIFICATION

June, 2023

Table of Contents

Definitions and Abbreviations.....	vii
1. PREREQUISITES.....	1
1.1 Network Connection to GePG.....	1
1.2 Billing System.....	1
1.3 API End Points and Call Back URLs.....	1
1.4 SMS Account.....	1
1.5 Digital Certificate.....	1
1.6 Message Exchange.....	1
1.7 Message Encoding.....	2
1.8 Special Characters.....	2
2. MESSAGE STRUCTURE.....	3
2.1 Request Message.....	3
2.2 Response Message.....	3
3. MESSAGE HANDLING.....	4
3.1 Message Exchange Mode.....	4
3.2 Protocol (HTTP/HTTPS).....	4
3.3 Message Acknowledgment and Response.....	4
3.4 Retry Mechanism.....	4
3.4.1 Institution to GePG.....	4
3.4.2 GePG to Institution.....	4
4. BILL POSTING.....	6
4.1 Bill Posting Information Flow.....	6
4.2 Bill Submission.....	7
4.2.1 Combined Bill Control Number.....	7
4.2.1.1 Combined Bill Control Number Request.....	8
4.2.1.2 Combined Bill Control Number Request Acknowledgment.....	10
a) Successful Acknowledgment.....	10
b) Failure Acknowledgment.....	10
4.2.1.3 Combined Bill Control Number Final Response.....	10
a) Successful Response.....	10
b) Failure Response.....	11
4.2.1.4 Combined Bill Control Number Final Response Acknowledgment.....	11
a) Successful Acknowledgment.....	11
b) Failure Acknowledgment.....	12

4.2.2 Normal Bill Control Number.....	12
4.2.2.1 Normal Bill Control Number Request.....	13
4.2.2.2 Normal Bill Control Number Final Response.....	14
a) Successful Response.....	14
b) Failure Response.....	14
4.3 Bill Control Number Reuse.....	15
4.3.1 Combined Bill Control Number Reuse.....	15
4.3.1.1 Combined Bill Control Number Reuse Request.....	16
4.3.2 Normal Bill Control Number Reuse.....	18
4.3.2.1 Normal Bill Control Number Reuse Request.....	18
4.4 Bill Change (Expiry Extension).....	19
4.4.1 Combined Bill Control Number Change.....	20
4.4.1.1 Combined Bill Control Number Change Request.....	20
4.4.2 Normal Bill Control Number Change.....	21
4.4.2.1 Normal Bill Control Number Change Request.....	21
4.5 Bill Cancellation.....	22
4.5.1 Combined Bill Control Number Cancellation Request.....	22
4.5.2 Normal Bill Control Number Cancellation Request.....	23
4.5.3 Bill Cancellation Response.....	24
4.6 Fields and parameters definition(s).....	24
5. QUOTE POSTING.....	30
5.1. Quote Information Posting Flow.....	30
5.2. Quote Information Notification.....	31
5.3. Quote Information Notification Acknowledgment.....	32
5.4. Quote Notification Response.....	32
5.5. Quote Notification Response Acknowledgment.....	33
6. PAYMENT POSTING.....	36
6.1. Payment Posting information Flow.....	36
6.2. Payment Information posting.....	37
6.3. Payment Information posting Acknowledgment.....	38
7. RECONCILIATION POSTING.....	41
7.1. Reconciliation Posting Information Flow.....	41
7.2. Reconciliation Request.....	42
7.3. Reconciliation Request Acknowledgment.....	42
7.4. Reconciliation Response.....	43
7.5. Reconciliation Response Acknowledgment.....	44

8. DIGITAL SIGNATURE SECURITY.....	48
8.1. Overview.....	48
8.2. Digital Signature.....	48
8.3. Digital Signature File Format.....	48
8.4. Signature Algorithm.....	48
8.5. Signed Final Payload Format.....	49
9. APPENDIX.....	50
9.1. Payer Identity Types.....	50
9.2. Response Codes.....	50

Document Description

Document Name	Government electronic Payment Gateway Programming Interface (API) for Service Provider (SP)
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Document version Control

Version	Date	Change Description
1.0	24-02-2017	Baseline initial version created.
1.1	24-02-2017	Added Token Request Procedure
1.2	25-02-2017	I. Ack XML simplified II. Added payment posting section
1.3	16-05-2017	Added Bill cancellation feature
1.4	19-07-2017	Update bill information
1.4.1	12-10-2017	- Added Exact bill definition - Added CtrAccNum tag in payment notification
2.0	28-10-2017	- Added digital signature - Added reconciliation message - Added data type definitions
2.0.1	31-10-2017	- Document formatting - Added reserved fields on recon message
2.0.2	18-01-2018	Added Signed Bill Cancellation Request Format
3.0	06-07-2018	- General Document Formatting - Added GePG online section - Added document pre-requisite section
4.0	01-07-2019	- Document Formatting - Added Bill control number reuse - Added Bill information update - Added Reason tag in Bill cancellation request
4.1	25-01-2020	- Document Formatting - Added Minimum Bill Payable Amount, Bill Payment Plan, Bill Payment Limitation Type, Bill Payment Limitation Amount, Collection

		Payment Service Provider
5.0	01-06-2023	General Improvements • Added Requests identification numbers • Added Quote Flow for Prepaid • Added Request Header for Encryption Algorithm • Added Support for Multiple Encryption Algorithms • Additional of tags ○ Online and normal payment notification ○ SP Reconciliation Normal Bill Request API • Submission of Bill Request with ○ Collection Centre ○ Bill amount Exchange Rate Bill Control Number Reuse API • Added Bill Control Number Reuse Reason • Allow Reuse of control number with less bill amount compared to existing balance Bill Information Update API • Bill Update Reason • Bill Update Approval User Bill Cancellation API • Allow cancellation of partially paid bill • Bill Cancellation Approval User Introduction of Combined Bill Control Number (Customer Control Number)

Definitions and Abbreviations

GePG	- Government electronic Payment Gateway
HTTP	- Hypertext Transfer Protocol
HTTPS	- Hypertext Transfer Protocol Secure
MNO	- Mobile Network Operator
PSP	- Payment Service Provider
SIT	- System Integration Test
SP	- Service Provider
UAT	- User Acceptance Test
VPN	- Virtual Private Network

1. PREREQUISITES

1.1 Network Connection to GePG

Integrating institution is supposed to establish secure Virtual Private Network (VPN) connection with GePG Data Center.

1.2 Billing System

Integrating institution must have a working or completely developed billing system that is ready for integration with GePG. The billing system shall be subjected to GePG System Integration Test (SIT) and User Acceptance Test (UAT); and once all pass then shall be allowed to migrate to GePG production environment.

1.3 API End Points and Call Back URLs

Integrating institution must provide the following URLs for receiving response or notification from GePG into her billing system: -

- Bill Submission Response
- Payment Notifications
- Successful Payments Summary Response
- Quote Request

1.4 SMS Account

Integrating institution may be required to provide SMS account approved by Government to be used in sending SMS for payment notification.

1.5 Digital Certificate

Integrating institution must share her public certificate(s) and obtain GePG public certificate(s) for validation of integrity and authenticity of messages exchanged between billing system and GePG. Expired or compromised certificates will be blacklisted and messages signed with such certificates will be rejected.

NOTE:

Testing certificate(s) may be provided by GePG and shall ONLY be used during integration testing sessions on GePG UAT environment.

1.6 Message Exchange

The transport protocol of this interface is XML over HTTP.

- a) The incoming messages from Billing System may arise from HTTP or HTTPS.
- b) The expected HTTP method is always POST.
- c) Endpoints shall be communicated during integration process. There shall be endpoints for testing environment and others for production environment.

1.7 Message Encoding

Message encoding shall use utf-8

1.8 Special Characters

Xml body content should ensure all the special characters are escaped.

Character	Description	Escaped by
&	Ampersand	&
<	Less-than	<
>	Greater-than	>
"	Quotes	"
'	Apostrophe	'

2. MESSAGE STRUCTURE

2.1 Request Message

GePG Request shall contain two (2) parts i.e HTTP Header and Payload in xml format.

HTTP header shall contain the following parameters when sending request to GePG.

Parameter	Values/Description
Content-Type	Application/xml
Gepg-Com	Shall be given during the integration
Gepg-Code	Shall be given during the integration
Gepg-Alg	Shall be given during the integration

Payload shall change based on service encountered (Please refer to respective section in this document).

2.2 Response Message

GePG Response shall contain two (2) parts i.e HTTP Header and Payload in xml format.

HTTP header shall contain the following parameters when responding to billing system.

Parameter	Values/Description
Content-Type	Application/xml

Payload shall change based on service encountered (Please refer to respective section in this document).

3. MESSAGE HANDLING

3.1 Message Exchange Mode

GePG APIs make use of asynchronous and synchronous modes of communication to complete requests processing. In asynchronous mode, GePG will receive requests from clients and acknowledge in the same session call. Later after request processing, GePG will call back clients system with request processing results at a callback url provided by client. In synchronous mode, GePG will receive requests, process them and return processing results in the same session call.

It is highly recommended for clients' systems to acknowledge to GePG callback requests immediately and proceed with other internal processing later.

3.2 Protocol (HTTP/HTTPS)

GePG APIs implements REST API Web Service where all requests will be received through http or https POST method.

3.3 Message Acknowledgment and Response

In every message exchange between GePG and Institution System, message will be acknowledged and/or responded according to each API specification. Acknowledgement and response from GePG shall have at least one GePG response code and its description. If there are multiple GePG response codes; contents will be separated by semicolon (;). Institution system acknowledgement or response shall contain one of the response codes. On receipt of the mentioned response code, GePG shall perform appropriate action as per received response code. Otherwise receipt of any other status code, GePG will assume message was received and processed successfully.

3.4 Retry Mechanism

3.4.1 Institution to GePG

Every API request from Institutions' systems to GePG will be acknowledged as per API specification. If an Institution does not receive request acknowledgement from GePG, Institution system has to retry sending an unacknowledged request provided that communication channel between the systems is working as agreed.

3.4.2 GePG to Institution

Either, GePG is going to follow the following guidelines to handle retry:-

a) No communication to Institution System

GePG will keep messages in waiting state, check for connectivity once in every hour and resume sending to destination system once connectivity is restored.

b) Connection to Institution System Times Out

GePG will keep retrying sending unacknowledged requests to Institutions' system after every half an hour for a period of twenty four hours (24) when pending unacknowledged requests will grouped into no connectivity messages.

c) Invalid acknowledgement content

GePG will attempt to send message to Institution system once in every half an hour for a period of twenty four hours when messages will be discarded.

4. BILL POSTING

4.1 Bill Posting Information Flow

- i. Integrating institution billing system composes bill submission request as per Bill Submission Request xml format and submit (single bill or multiple bills at once) to GePG, until GePG acknowledge receiving.
- ii. GePG receive the bill, immediately send back the acknowledgement as per Bill Submission Request Acknowledgement xml format to the billing system.
- iii. GePG process the bill, compose the bill response as per Bill Submission Response xml format and send to a billing system until acknowledge it. Billing system must acknowledge response before doing any internal processing.
- iv. The billing system must acknowledge before doing any internal processing to a bill response as per Bill Submission Response Acknowledgement xml format.

NOTE:

1. In case billing system does not receive acknowledgement from GePG in step ii. above, it must keep retrying until acknowledgment is received. Unless there is a connectivity problem between the two systems or any other problem.
2. In case billing system receives acknowledgement with response code other than **7101** in step ii. above, it should not perform retry until the returned code is acted accordingly. For example, response code **7242** implies invalid xml schema tag or data, therefore retry can only be done after rectification of xml or data that resulted to invalid xml.
3. In case the GePG does not receive the acknowledgment or receive invalid xml format from billing system in step iv. above, it will keep on retrying to send the bill results until the respective request aged 24 hours at which GePG will discard message.
4. After the message has been discarded by GePG, the institution billing system will be required to resubmit the same request (i. above) in order to retrieve the message.
5. Resubmitted bill request (bill request with the same bill identification) will be responded with the previously issued bill control number and response code **7226** indicating a duplicate request.

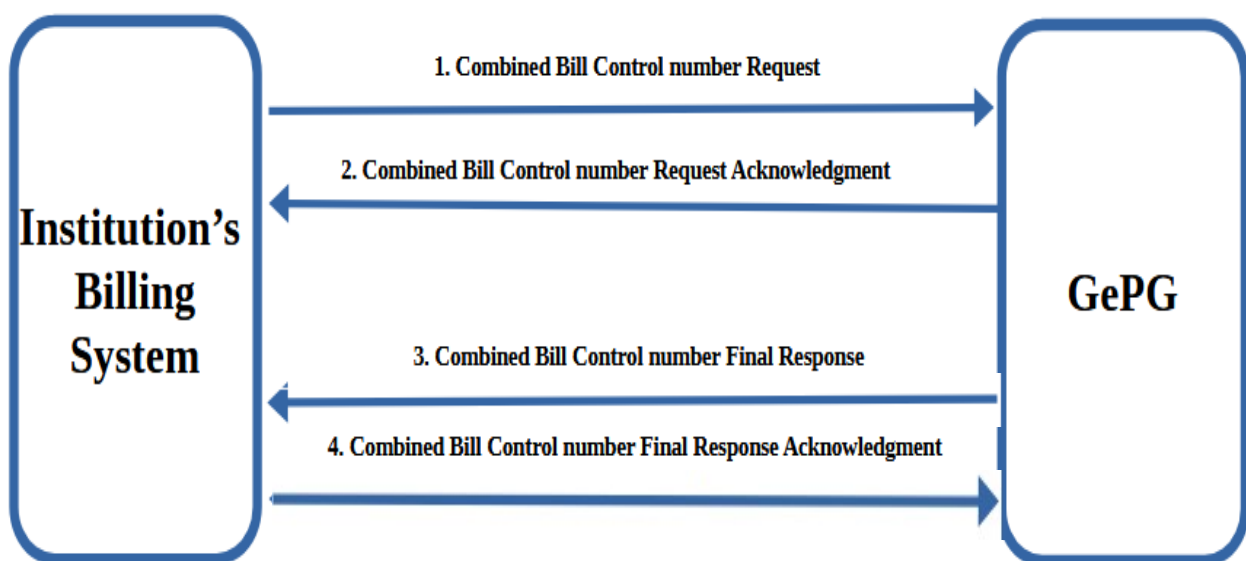
4.2 Bill Submission

4.2.1 Combined Bill Control Number

Combined Bill Control Number (customer bill control number) is a type of bill control number that is issued to represent other bill control numbers. The combined bill control numbers may come from single government institution or come from different institutions owing the same customer. The different normal bill control numbers are combined to give the customer a convenient way of completing the process of payment of several bills by using one reference.

Rules and things to consider in Combined Bill Control Number

- a) If combined bills are from different institutions, there has to be an understanding amongst them on revenue sources, billing departments and customer identification to be used in combined bills and which system will be responsible in submitting bill requests to GePG.
- b) Combined bills will share billing currency and payment option shall be EXACT
- c) Each combined bill may have multiple charged items, and each item must show the originating bill by providing the original bill number(**BillId**) in the field **RefBillId**.
- d) Each combined bill may specify its own expiry date but GePG will consider the earliest expiry time as the expiry time for the combine bill.
- e) GePG will process the bill as a whole and return the generated Customer Bill Control Number and corresponding Bill Control Numbers in a single response.



4.2.1.1 Combined Bill Control Number Request

```
<Gepg>
  <billSubReq>
    <BillHdr>
      <ReqId>SP20210205130219</ReqId>
      <SpGrpCode>SPG1000</SpGrpCode>
      <SysCode>SYS001</SysCode>
      <BillTyp>2</BillTyp>
      <PayTyp>2</PayTyp>
      <GrpBillId>PRI1581953689</GrpBillId>
    </BillHdr>
    <BillDtls>
      <BillDtl>
        <BillId>SP1158195368931</BillId>
        <SpCode>SP99128</SpCode>
        <CollCentCode>14476430002</CollCentCode>
        <BillDesc>Purchase of Timber</BillDesc>
        <CustTin>111535345</CustTin>
        <CustId>UB132424114</CustId>
        <CustIdTyp>1</CustIdTyp>
        <CustAcct>T123ABC</CustAcct>
        <CustName>Ubuntu Wamwana</CustName>
        <CustCellNum>255713525539</CustCellNum>
        <CustEmail>payer@yahoo.com</CustEmail>
        <BillGenDt>2018-10-22T10:00:00</BillGenDt>
        <BillExprDt>2023-10-30T10:00:30</BillExprDt>
        <BillGenBy>Allen Scott</BillGenBy>
        <BillApprBy>John Legend</BillApprBy>
        <BillAmt>30000.00</BillAmt>
        <BillEqvAmt>30000.00</BillEqvAmt>
        <MinPayAmt>0.01</MinPayAmt>
        <Ccy>TZS</Ccy>
        <ExchRate>1.00</ExchRate>
        <BillPayOpt>1</BillPayOpt>
        <PayPlan>1</PayPlan>
        <PayLimTyp>1</PayLimTyp>
        <PayLimAmt>0.00</PayLimAmt>
        <CollPsp></CollPsp>
        <BillItems>
          <BillItem>
            <RefBillId>SP11581953689</RefBillId>
            <SubSpCode>3001</SubSpCode>
            <GfsCode>1403541</GfsCode>
            <BillItemRef>B1IT115800000</BillItemRef>
            <UseItemRefOnPay>N</UseItemRefOnPay>
            <BillItemAmt>10000.00</BillItemAmt>
            <BillItemEqvAmt>10000.00</BillItemEqvAmt>
            <CollSp>SP99128</CollSp>
          </BillItem>
          <BillItem>
            <RefBillId>SP11581953689</RefBillId>
            <SubSpCode>3001</SubSpCode>
            <GfsCode>1403542</GfsCode>
            <BillItemRef>B1IT11581AJ</BillItemRef>
            <UseItemRefOnPay>N</UseItemRefOnPay>
            <BillItemAmt>20000.00</BillItemAmt>
            <BillItemEqvAmt>20000.00</BillItemEqvAmt>
            <CollSp>SP99128</CollSp>
          </BillItem>
        </BillItems>
      </BillDtl>
    </BillDtls>
  </billSubReq>
</Gepg>
```

```

        </BillItem>
    </BillItems>
</BillDtl>
<BillDtl>
    <BillId>12230005031</BillId>
    <SpCode>SP99108</SpCode>
    <CollCentCode>14476430002</CollCentCode>
    <BillDesc>Taxes on timber services</BillDesc>
    <CustTin>111535345</CustTin>
    <CustId>UB132424114</CustId>
    <CustIdTyp>1</CustIdTyp>
    <CustAcct>34233FFE</CustAcct>
    <CustName>Ubuntu Wamwana</CustName>
    <CustCellNum>255713525539</CustCellNum>
    <CustEmail>payer@yahoo.com</CustEmail>
    <BillGenDt>2018-10-22T10:00:00</BillGenDt>
    <BillExprDt>2023-10-30T10:00:30</BillExprDt>
    <BillGenBy>Pamba Were</BillGenBy>
    <BillApprBy>John Buyu</BillApprBy>
    <BillAmt>53600.00</BillAmt>
    <BillEqvAmt>53600.00</BillEqvAmt>
    <MinPayAmt>0.01</MinPayAmt>
    <Ccy>TZS</Ccy>
    <ExchRate>1.00</ExchRate>
    <BillPayOpt>1</BillPayOpt>
    <PayPlan>1</PayPlan>
    <PayLimTyp>1</PayLimTyp>
    <PayLimAmt>0.00</PayLimAmt>
    <CollPsp></CollPsp>
    <BillItems>
        <BillItem>
            <RefBillId>122300050</RefBillId>
            <SubSpCode>2001</SubSpCode>
            <GfsCode>140206</GfsCode>
            <BillItemRef>B1IT11581953689</BillItemRef>
            <UseItemRefOnPay>N</UseItemRefOnPay>
            <BillItemAmt>50000.00</BillItemAmt>
            <BillItemEqvAmt>50000.00</BillItemEqvAmt>
            <CollSp>SP99842</CollSp>
        </BillItem>
        <BillItem>
            <RefBillId>SP11581953689</RefBillId>
            <SubSpCode>2001</SubSpCode>
            <GfsCode>140368</GfsCode>
            <BillItemRef>B1IT115819523</BillItemRef>
            <UseItemRefOnPay>N</UseItemRefOnPay>
            <BillItemAmt>3600.00</BillItemAmt>
            <BillItemEqvAmt>3600.00</BillItemEqvAmt>
            <CollSp>SP99128</CollSp>
        </BillItem>
    </BillItems>
</BillDtl>
</BillDtls>
</billSubReq>
<signature>SignatureGoesHere</signature>
</Gepg>

```


4.2.1.2 Combined Bill Control Number Request Acknowledgment

a) Successful Acknowledgment

```
<Gepg>
  <billSubReqAck>
    <AckId>GW20210205130219</AckId>
    <ReqId>SP20210205130219</ReqId>
    <AckStsCode>7101</AckStsCode>
    <AckStsDesc>Successful</AckStsDesc>
  </billSubReqAck>
  <signature>SignatureGoesHere</signature>
</Gepg>
```

b) Failure Acknowledgment

```
<Gepg>
  <billSubReqAck>
    <AckId>GW20210205130219</AckId>
    <ReqId>SP20210205130219</ReqId>
    <AckStsCode>7242</AckStsCode>
    <AckStsDesc>Invalid Request Data</AckStsDesc>
  </billSubReqAck>
  <signature>SignatureGoesHere</signature>
</Gepg>
```

4.2.1.3 Combined Bill Control Number Final Response

a) Successful Response

```
<Gepg>
  <billSubRes>
    <BillHdr>
      <ResId>GW20210205130219</ResId>
      <ReqId>SP20210205130219</ReqId>
      <GrpBillId>PRI1581953689</GrpBillId>
      <CustCntrNum>999910859999771</CustCntrNum>
      <ResSts>GS</ResSts>
      <ResStsCode>7101</ResStsCode>
      <ResStsDesc>Successful</ResStsDesc>
    </BillHdr>
    <BillDtls>
      <BillDtl>
        <BillId>B11581953689</BillId>
        <BillCntrNum>991280104006</BillCntrNum>
        <BillStsCode>7101</BillStsCode>
        <BillStsDesc>Successful</BillStsDesc>
      </BillDtl>
      <BillDtl>
        <BillId>B21581953689</BillId>
        <BillCntrNum>999910859999770</BillCntrNum>
        <BillStsCode>7101</BillStsCode>
        <BillStsDesc>Successful</BillStsDesc>
      </BillDtl>
    </BillDtls>
  </billSubRes>
  <signature>SignatureGoesHere</signature>
</Gepg>
```

b) Failure Response

```
<Gepg>
  <billSubRes>
    <BillHdr>
      <ResId>GW20210205130219</ResId>
      <ReqId>SP20210205130219</ReqId>
      <GrpBillId>PRI1581953689</GrpBillId>
      <CustCntrNum>0</CustCntrNum>
      <ResSts>GF</ResSts>
      <ResStsCode>7102</ResStsCode>
      <ResStsDesc>Failure</ResStsDesc>
    </BillHdr>
    <BillDtls>
      <BillDtl>
        <BillId>B11581955043</BillId>
        <BillCntrNum>0</BillCntrNum>
        <BillStsCode>7101</BillStsCode>
        <BillStsDesc>Successful</BillStsDesc>
      </BillDtl>
      <BillDtl>
        <BillId>B11581955043</BillId>
        <BillCntrNum>0</BillCntrNum>
        <BillStsCode>7223</BillStsCode>
        <BillStsDesc>Invalid or inactive government financial statistic code(gfs) or service type</BillStsDesc>
      </BillDtl>
    </BillDtls>
  </billSubRes>
  <signature>SignatureGoesHere</signature>
</Gepg>
```

4.2.1.4 Combined Bill Control Number Final Response Acknowledgment

a) Successful Acknowledgment

```
<Gepg>
  <billSubResAck>
    <AckId>SP20210205130219</AckId>
    <ResId>GW20210205130219</ResId>
    <AckStsCode>7101</AckStsCode>
  </billSubResAck>
  <signature>SignatureGoesHere</signature>
</Gepg>
```

b) Failure Acknowledgment

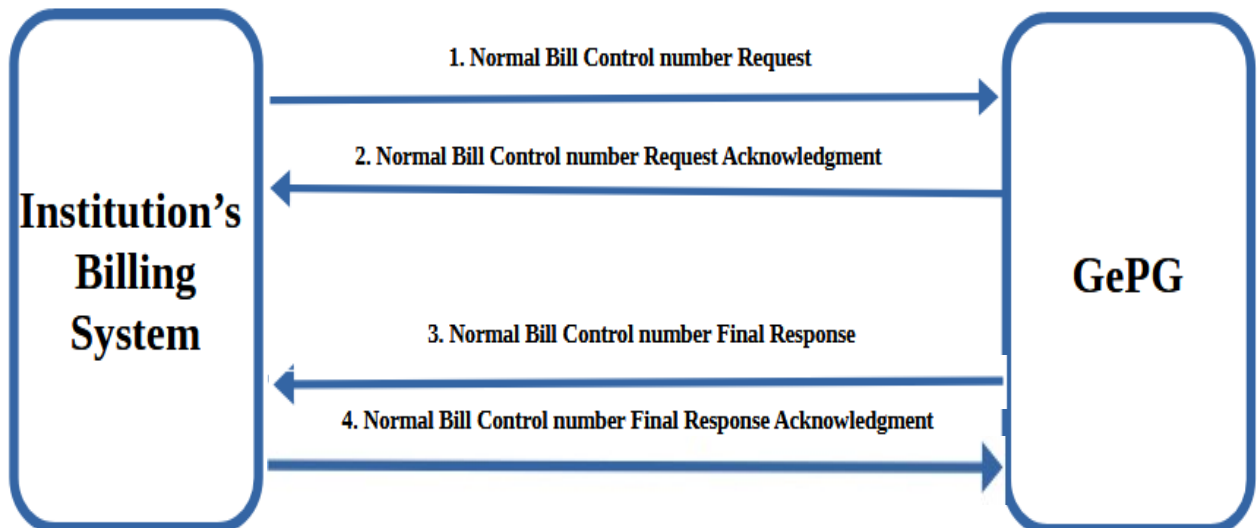
```
<Gepg>
  <billSubResAck>
    <AckId>SP20210205130219</AckId>
    <ResId>GW20210205130219</ResId>
    <AckStsCode>7303</AckStsCode>
  </billSubResAck>
  <signature>SignatureGoesHere</signature>
</Gepg>
```

4.2.2 Normal Bill Control Number

Normal Bill Control Number is a type of bill control number that is issued for a single bill. This control number request may contain multiple bills but each bill is independent and may be issued to the same customer or different customers. Each bill in normal bill control number request is processed independently and responses are posted to the requesting system independently.

Upon receipt and preliminary bill validations of normal bill control number request, GePG will acknowledge to the posting system with successful or unsuccessful acknowledgement.

Either, after processing completion GePG will send bill processing results to the institution billing system. The Institution billing system has to acknowledge to the response received.



4.2.2.1 Normal Bill Control Number Request

```
<Gepg>
  <billSubReq>
    <BillHdr>
      <ReqId>SP20210205130219</ReqId>
      <SpGrpCode>SP99128</SpGrpCode>
      <SysCode>SYS001</SysCode>
      <BillTyp>1</BillTyp>
      <PayTyp>1</PayTyp>
      <GrpBillId>SP1158195368931</GrpBillId>
    </BillHdr>
    <BillDtls>
      <BillDtl>
        <BillId>SP1158195368931</BillId>
        <SpCode>SP99128</SpCode>
        <CollCentCode>14476430002</CollCentCode>
        <BillDesc>Purchase of Timber</BillDesc>
        <CustTin>111535345</CustTin>
        <CustId>UB132424114</CustId>
        <CustIdTyp>1</CustIdTyp>
        <CustAcct>T123ABC</CustAcct>
        <CustName>Ubuntu Wamwana</CustName>
        <CustCellNum>255713525539</CustCellNum>
        <CustEmail>payer@yahoo.com</CustEmail>
        <BillGenDt>2018-10-22T10:00:00</BillGenDt>
        <BillExprDt>2023-10-30T10:00:30</BillExprDt>
        <BillGenBy>Allen Scott</BillGenBy>
        <BillApprBy>John Legend</BillApprBy>
        <BillAmt>30000.00</BillAmt>
        <BillEqvAmt>30000.00</BillEqvAmt>
        <MinPayAmt>0.01</MinPayAmt>
        <Ccy>TZS</Ccy>
        <ExchRate>1.00</ExchRate>
        <BillPayOpt>1</BillPayOpt>
        <PayPlan>1</PayPlan>
        <PayLimTyp>1</PayLimTyp>
        <PayLimAmt>0.00</PayLimAmt>
        <CollPsp></CollPsp>
        <BillItems>
          <BillItem>
            <RefBillId>SP11581953689</RefBillId>
            <SubSpCode>3001</SubSpCode>
            <GfsCode>1403541</GfsCode>
            <BillItemRef>B1IT115800000</BillItemRef>
            <UseItemRefOnPay>N</UseItemRefOnPay>
            <BillItemAmt>10000.00</BillItemAmt>
            <BillItemEqvAmt>10000.00</BillItemEqvAmt>
            <CollSp>SP99128</CollSp>
          </BillItem>
          <BillItem>
            <RefBillId>SP11581953689</RefBillId>
            <SubSpCode>3001</SubSpCode>
            <GfsCode>1403542</GfsCode>
            <BillItemRef>B1IT11581AJ</BillItemRef>
            <UseItemRefOnPay>N</UseItemRefOnPay>
            <BillItemAmt>20000.00</BillItemAmt>
```

```

        <BillItemEqvAmt>20000.00</BillItemEqvAmt>
        <CollSp>SP99128</CollSp>
    </BillItem>
</BillItems>
</BillDtl>
</BillDtls>
</billSubReq>
<signature>SignatureGoesHere</signature>
</Gepg>

```

4.2.2.2 Normal Bill Control Number Final Response

a) Successful Response

```

<Gepg>
  <billSubRes>
    <BillHdr>
      <ResId>GW20210205130219</ResId>
      <ReqId>SP20210205130219</ReqId>
      <GrpBillId>SP1158195368931</GrpBillId>
      <CustCntrNum>991280104006</CustCntrNum>
      <ResSts>GS</ResSts>
      <ResStsCode>7101</ResStsCode>
      <ResStsDesc>Successful</ResStsDesc>
    </BillHdr>
    <BillDtls>
      <BillDtl>
        <BillId>SP1158195368931</BillId>
        <BillCntrNum>991280104006</BillCntrNum>
        <BillStsCode>7101</BillStsCode>
        <BillStsDesc>Successful</BillStsDesc>
      </BillDtl>
    </BillDtls>
  </billSubRes>
  <signature>SignatureGoesHere</signature>
</Gepg>

```

b) Failure Response

```

<Gepg>
  <billSubRes>
    <BillHdr>
      <ResId>GW20210205130219</ResId>
      <ReqId>SP20210205130219</ReqId>
      <GrpBillId>SP1158195368931</GrpBillId>
      <CustCntrNum>0</CustCntrNum>
      <ResSts>GF</ResSts>
      <ResStsCode>7102</ResStsCode>
      <ResStsDesc>Failure</ResStsDesc>
    </BillHdr>
    <BillDtls>
      <BillDtl>
        <BillId>SP1158195368931</BillId>
        <BillCntrNum>0</BillCntrNum>
        <BillStsCode>7223</BillStsCode>
        <BillStsDesc>Invalid or inactive
government financial statistic's
code</BillStsDesc>
      </BillDtl>
    </BillDtls>
  </billSubRes>
  <signature>SignatureGoesHere</signature>
</Gepg>

```

```

        </BillDtl>
      </BillDtls>
    </billSubRes>
    <signature>SignatureGoesHere</signature>
  </Gepg>

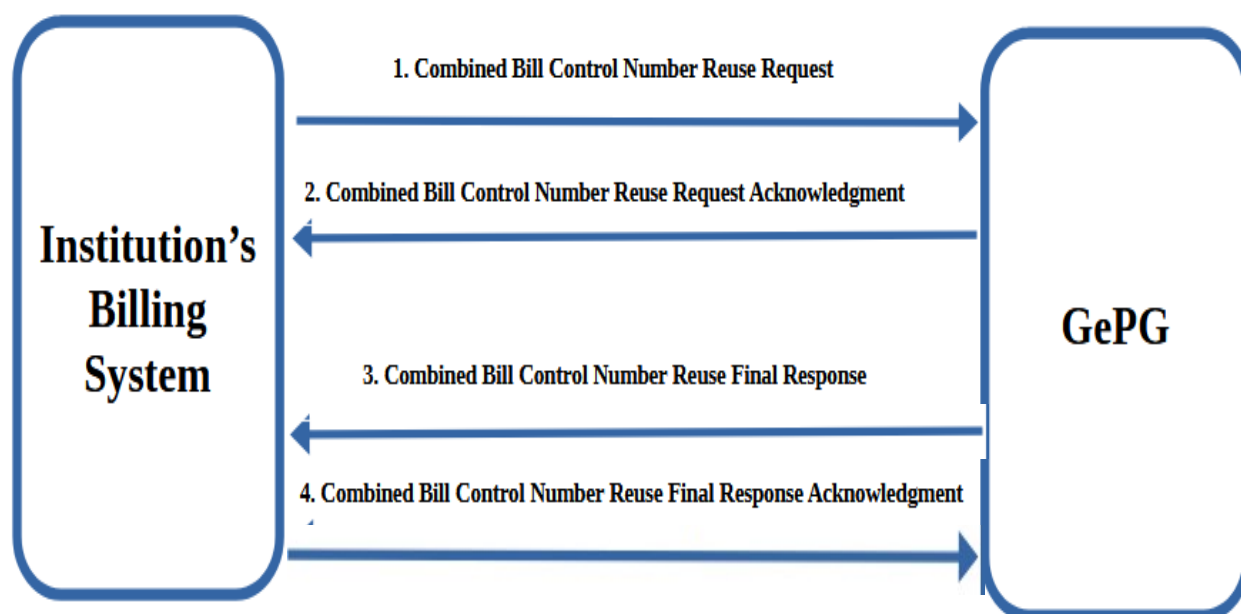
```

4.3 Bill Control Number Reuse

Bill Control Number Reuse is a feature that enables the same control number issued earlier to be reused again. The reused control number has to be submitted together with the bill details that may have changed since previous submission. If the previous issued control number had not expired yet or the billed amount has decreased, the new submitted request has to include a reason for reuse.

Upon receipt and preliminary bill validations of bill control number reuse request, GePG will acknowledge to the posting system with successful or unsuccessful acknowledgement.

4.3.1 Combined Bill Control Number Reuse



4.3.1.1 Combined Bill Control Number Reuse Request

Final bill control number response and final response acknowledgement for combined reuse request is the same as normal combined control number request.

```
<Gepg>
  <billSubReq>
    <BillHdr>
      <ReqId>SP20210205130219</ReqId>
      <SpGrpCode>SPG1000</SpGrpCode>
      <SysCode>SYS001</SysCode>
      <BillTyp>2</BillTyp>
      <PayTyp>2</PayTyp>
      <GrpBillId>PR11581953689</GrpBillId>
      <CustCntrNum>999910859999771</CustCntrNum>
    </BillHdr>
    <BillDtls>
      <BillDtl>
        <BillId>SP1158195368931</BillId>
        <SpCode>SP99128</SpCode>
        <CollCentCode>14476430002</CollCentCode>
        <BillDesc>Purchase of Timber</BillDesc>
        <CustTin>111535345</CustTin>
        <CustId>UB132424114</CustId>
        <CustIdTyp>1</CustIdTyp>
        <CustAcct>T123ABC</CustAcct>
        <CustName>Ubuntu Wamwana</CustName>
        <CustCellNum>255713525539</CustCellNum>
        <CustEmail>payer@yahoo.com</CustEmail>
        <BillGenDt>2018-10-22T10:00:00</BillGenDt>
        <BillExprDt>2023-10-30T10:00:30</BillExprDt>
        <BillGenBy>Allen Scott</BillGenBy>
        <BillApprBy>John Legend</BillApprBy>
        <BillAmt>30000.00</BillAmt>
        <BillEqvAmt>30000.00</BillEqvAmt>
        <MinPayAmt>0.01</MinPayAmt>
        <Ccy>TZS</Ccy>
        <ExchRate>1.00</ExchRate>
        <BillPayOpt>1</BillPayOpt>
        <PayPlan>1</PayPlan>
        <PayLimTyp>1</PayLimTyp>
        <PayLimAmt>0.00</PayLimAmt>
        <CollPsp></CollPsp>
        <BillCntrNum>991280104006</BillCntrNum>
        <ReuseReasn>Billed amount adjustment</ReuseReasn>
        <BillItems>
          <BillItem>
            <RefBillId>SP11581953689</RefBillId>
            <SubSpCode>3001</SubSpCode>
            <GfsCode>1403541</GfsCode>
            <BillItemRef>B1IT115800000</BillItemRef>
            <UseItemRefOnPay>N</UseItemRefOnPay>
            <BillItemAmt>10000.00</BillItemAmt>
            <BillItemEqvAmt>10000.00</BillItemEqvAmt>
            <CollSp>SP99128</CollSp>
          </BillItem>
          <BillItem>
            <RefBillId>SP11581953689</RefBillId>
            <SubSpCode>3001</SubSpCode>
            <GfsCode>1403542</GfsCode>
```

```

        <BillItemRef>B1IT11581AJ</BillItemRef>
        <UseItemRefOnPay>N</UseItemRefOnPay>
        <BillItemAmt>20000.00</BillItemAmt>
        <BillItemEqvAmt>20000.00</BillItemEqvAmt>
        <CollSp>SP99128</CollSp>
    </BillItem>
</BillItems>
</BillDtl>
<BillDtl>
    <BillId>12230005031</BillId>
    <SpCode>SP99108</SpCode>
    <CollCentCode>14476430002</CollCentCode>
    <BillDesc>Taxes on timber services</BillDesc>
    <CustTin>111535345</CustTin>
    <CustId>UB132424114</CustId>
    <CustIdTyp>1</CustIdTyp>
    <CustAcct>34233FFE</CustAcct>
    <CustName>Ubuntu Wamwana</CustName>
    <CustCellNum>255713525539</CustCellNum>
    <CustEmail>payer@yahoo.com</CustEmail>
    <BillGenDt>2018-10-22T10:00:00</BillGenDt>
    <BillExprDt>2023-10-30T10:00:30</BillExprDt>
    <BillGenBy>Pamba Were</BillGenBy>
    <BillApprBy>John Buyu</BillApprBy>
    <BillAmt>53600.00</BillAmt>
    <BillEqvAmt>53600.00</BillEqvAmt>
    <MinPayAmt>0.01</MinPayAmt>
    <Ccy>TZS</Ccy>
    <ExchRate>1.00</ExchRate>
    <BillPayOpt>1</BillPayOpt>
    <PayPlan>1</PayPlan>
    <PayLimTyp>1</PayLimTyp>
    <PayLimAmt>0.00</PayLimAmt>
    <CollPsp></CollPsp>
    <BillCntrNum>999910859999770</BillCntrNum>
    <ReuseReasn>Billed amount adjustment</ReuseReasn>
    <BillItems>
        <BillItem>
            <RefBillId>122300050</RefBillId>
            <SubSpCode>2001</SubSpCode>
            <GfsCode>140206</GfsCode>
            <BillItemRef>B1IT11581953689</BillItemRef>
            <UseItemRefOnPay>N</UseItemRefOnPay>
            <BillItemAmt>50000.00</BillItemAmt>
            <BillItemEqvAmt>50000.00</BillItemEqvAmt>
            <CollSp>SP99842</CollSp>
        </BillItem>
        <BillItem>
            <RefBillId>SP11581953689</RefBillId>
            <SubSpCode>2001</SubSpCode>
            <GfsCode>140368</GfsCode>
            <BillItemRef>B1IT115819523</BillItemRef>
            <UseItemRefOnPay>N</UseItemRefOnPay>
            <BillItemAmt>3600.00</BillItemAmt>
            <BillItemEqvAmt>3600.00</BillItemEqvAmt>
            <CollSp>SP99128</CollSp>
        </BillItem>
    </BillItems>
</BillDtl>
</BillDtls>
</billSubReq>

```

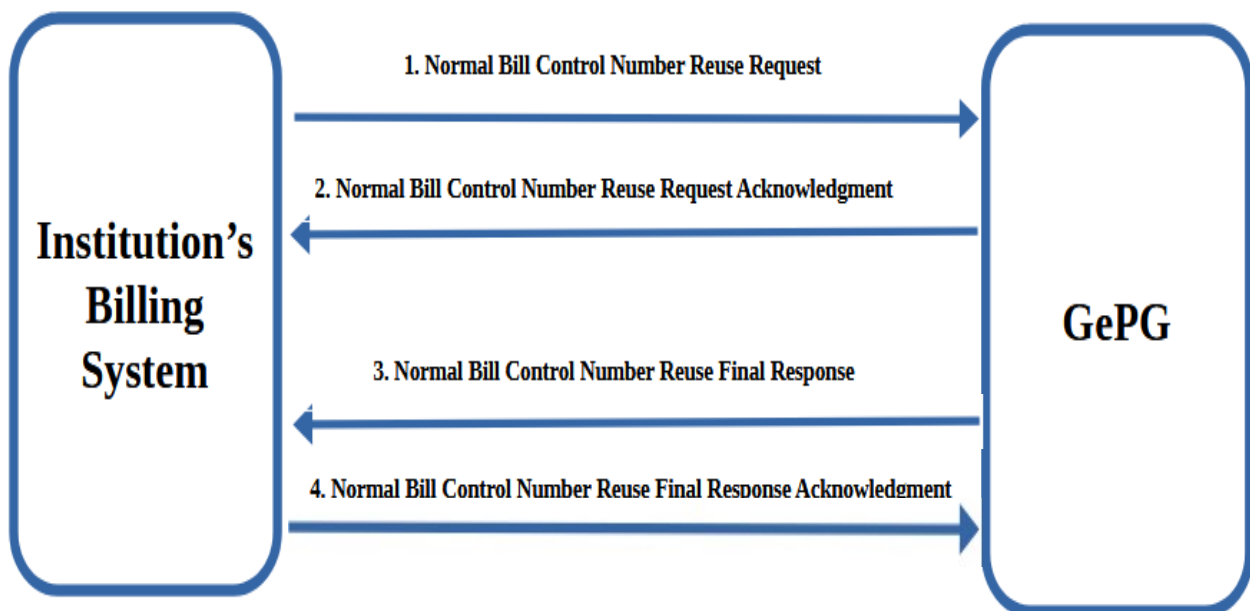


```
<signature>SignatureGoesHere</signature>
</Gepg>
```

4.3.2 Normal Bill Control Number Reuse

Normal Bill Control Number reuse API works as the normal bill control number request. Reuse request requires the previous issued bill control number to be included as one of the bill request parameters together with the reuse reason in case when the billed amount has changed or the previous issued normal bill control number has not expired yet.

GePG will ack to this request after receipt and preliminary validations. Final response from GePG to Institution billing system has to be acknowledged.



4.3.2.1 Normal Bill Control Number Reuse Request

```
<Gepg>
  <billSubReq>
    <BillHdr>
      <ReqId>SP20210205130219</ReqId>
      <SpGrpCode>SP99128</SpGrpCode>
      <SysCode>SYS001</SysCode>
      <BillTyp>1</BillTyp>
      <PayTyp>1</PayTyp>
      <GrpBillId>SP1158195368931</GrpBillId>
    </BillHdr>
    <BillDtls>
      <BillDtl>
        <BillId>SP1158195368931</BillId>
        <SpCode>SP99128</SpCode>
        <CollCentCode>14476430002</CollCentCode>
        <BillDesc>Purchase of Timber</BillDesc>
        <CustTin>111535345</CustTin>
      </BillDtl>
    </BillDtls>
  </billSubReq>
</Gepg>
```

```

<CustId>UB132424114</CustId>
<CustIdTyp>1</CustIdTyp>
<CustAcct>T123ABC</CustAcct>
<CustName>Ubuntu Wamwana</CustName>
<CustCellNum>255713525539</CustCellNum>
<CustEmail>payer@yahoo.com</CustEmail>
<BillGenDt>2018-10-22T10:00:00</BillGenDt>
<BillExprDt>2023-10-30T10:00:30</BillExprDt>
<BillGenBy>Allen Scott</BillGenBy>
<BillApprBy>John Legend</BillApprBy>
<BillAmt>30000.00</BillAmt>
<BillEqvAmt>30000.00</BillEqvAmt>
<MinPayAmt>0.01</MinPayAmt>
<Ccy>TZS</Ccy>
<ExchRate>1.00</ExchRate>
<BillPayOpt>1</BillPayOpt>
<PayPlan>1</PayPlan>
<PayLimTyp>1</PayLimTyp>
<PayLimAmt>0.00</PayLimAmt>
<CollPsp></CollPsp>
<BillCntrNum>9912859999770</BillCntrNum>
<ReuseReasn>Billed amount adjustment</ReuseReasn>
<BillItems>
  <BillItem>
    <RefBillId>SP11581953689</RefBillId>
    <SubSpCode>3001</SubSpCode>
    <GfsCode>1403541</GfsCode>
    <BillItemRef>B1IT115800000</BillItemRef>
    <UseItemRefOnPay>N</UseItemRefOnPay>
    <BillItemAmt>10000.00</BillItemAmt>
    <BillItemEqvAmt>10000.00</BillItemEqvAmt>
    <CollSp>SP99128</CollSp>
  </BillItem>
  <BillItem>
    <RefBillId>SP11581953689</RefBillId>
    <SubSpCode>3001</SubSpCode>
    <GfsCode>1403542</GfsCode>
    <BillItemRef>B1IT11581AJ</BillItemRef>
    <UseItemRefOnPay>N</UseItemRefOnPay>
    <BillItemAmt>20000.00</BillItemAmt>
    <BillItemEqvAmt>20000.00</BillItemEqvAmt>
    <CollSp>SP99128</CollSp>
  </BillItem>
</BillItems>
</BillDtl>
</BillDtls>
</billSubReq>
<signature>SignatureGoesHere</signature>
</Gepg>

```

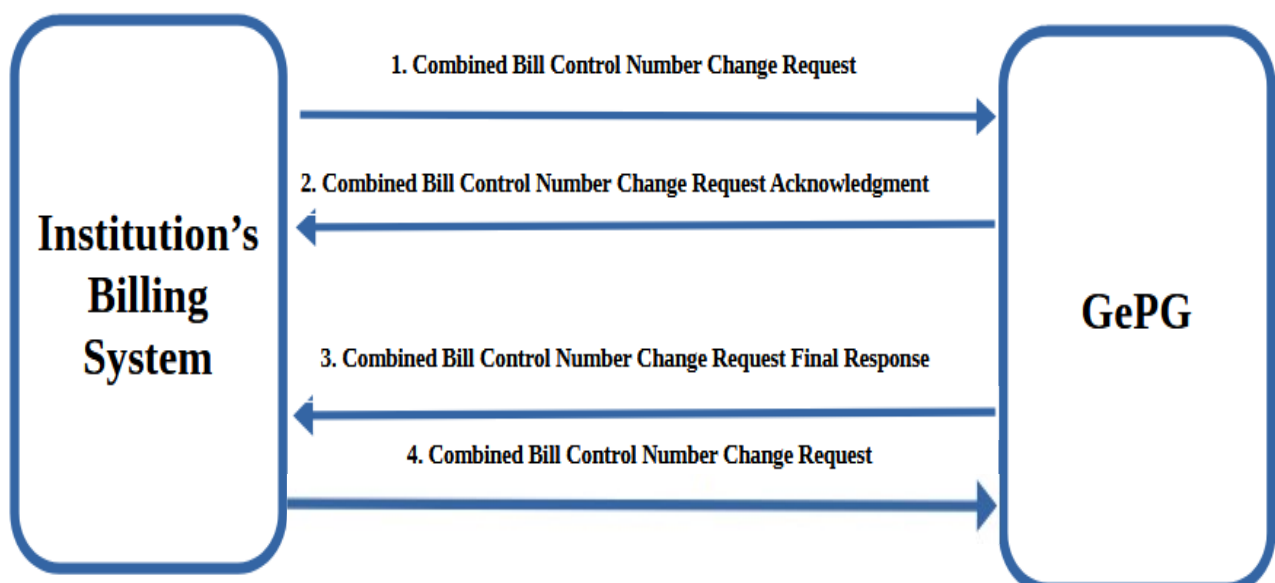
4.4 Bill Change (Expiry Extension)

Bill change or bill expiry extension API allows Institutions to extend pending bills expiry date to future times when needed. It requires submission of the original BillId, identifying the previous submitted bill, new expiry date and some other bill fields necessary for the update process.

Request Acknowledgements and Responses to bill expiry update remain the same as shown in combined bill control number request and normal bill control number request.

The Institution billing system has to acknowledge to the final response received with message schema.

4.4.1 Combined Bill Control Number Change



4.4.1.1 Combined Bill Control Number Change Request

```

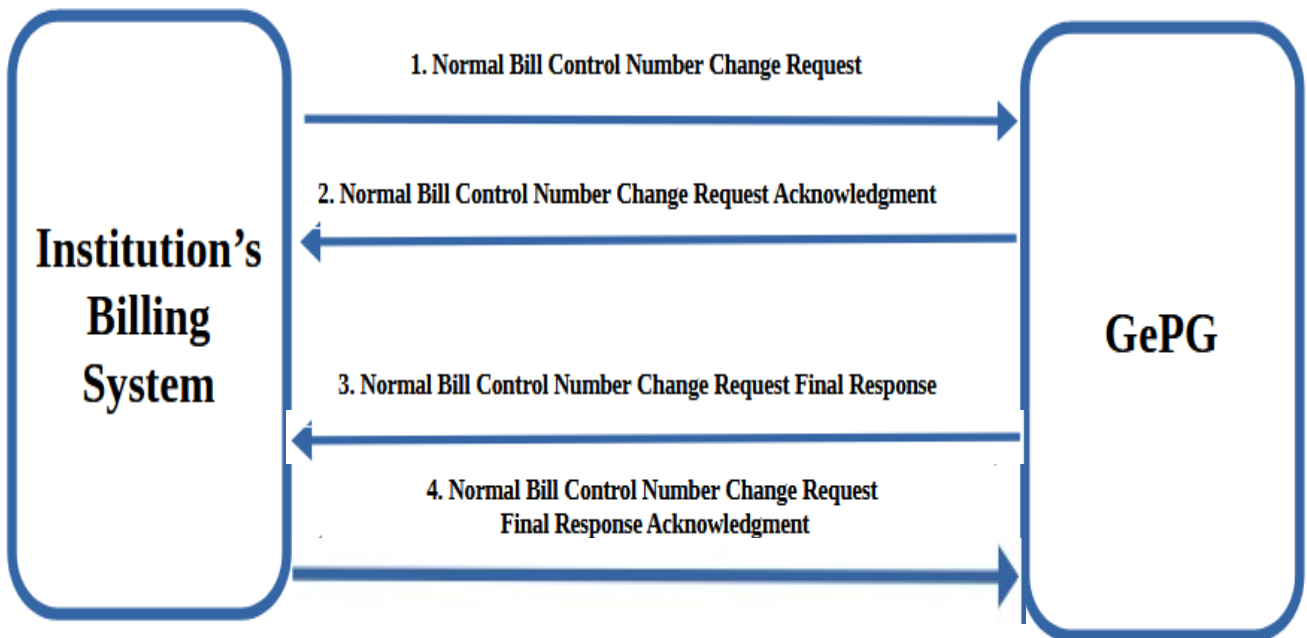
<Gepg>
  <billSubReq>
    <BillHdr>
      <ReqId>SP20210205130219</ReqId>
      <SpGrpCode>SPG1000</SpGrpCode>
      <SysCode>SYS001</SysCode>
      <BillTyp>2</BillTyp>
      <PayTyp>2</PayTyp>
      <GrpBillId>PRI1581953689</GrpBillId>
    </BillHdr>
    <BillDtls>
      <BillDtl>
        <BillId>SP1158195368931</BillId>
        <SpCode>SP99128</SpCode>
        <BillExprDt>2023-10-30T10:00:30</BillExprDt>
        <BillGenBy>Allen Scott</BillGenBy>
        <BillApprBy>John Legend</BillApprBy>
      </BillDtl>
    </BillDtls>
  </billSubReq>
</Gepg>
  
```

```

        <UpdateReasn>Time eextension after flood
        disasters</UpdateReasn>
    </BillDtl>
    <BillDtl>
        <BillId>12230005031</BillId>
        <SpCode>SP99108</SpCode>
        <BillExprDt>2023-10-30T10:00:30</BillExprDt>
        <BillGenBy>Pamba Were</BillGenBy>
        <BillApprBy>John Buyu</BillApprBy>
        <UpdateReasn>Time eextension after flood
        disasters</UpdateReasn>
    </BillDtl>
    </BillDtls>
</billSubReq>
<signature>SignatureGoesHere</signature>
</Gepg>

```

4.4.2 Normal Bill Control Number Change



4.4.2.1 Normal Bill Control Number Change Request

```

<Gepg>
    <billSubReq>
        <BillHdr>
            <ReqId>SP20210205130219</ReqId>
            <SpGrpCode>SPG1000</SpGrpCode>
            <SysCode>SYS001</SysCode>
            <BillTyp>1</BillTyp>
            <PayTyp>1</PayTyp>
            <GrpBillId>SP1158195368931</GrpBillId>
        </BillHdr>
        <BillDtls>
            <BillDtl>
                <BillId>SP1158195368931</BillId>
            </BillDtl>
        </BillDtls>
    </billSubReq>
</Gepg>

```

```

        <SpCode>SP99128</SpCode>
        <BillExprDt>2023-10-30T10:00:30</BillExprDt>
        <BillGenBy>Allen Scott</BillGenBy>
        <BillApprBy>John Legend</BillApprBy>
        <UpdateReasn>Time eextension after flood
        disasters</UpdateReasn>
    </BillDtl>
</BillDtls>
</billSubReq>
<signature>SignatureGoesHere</signature>
</Gepg>

```

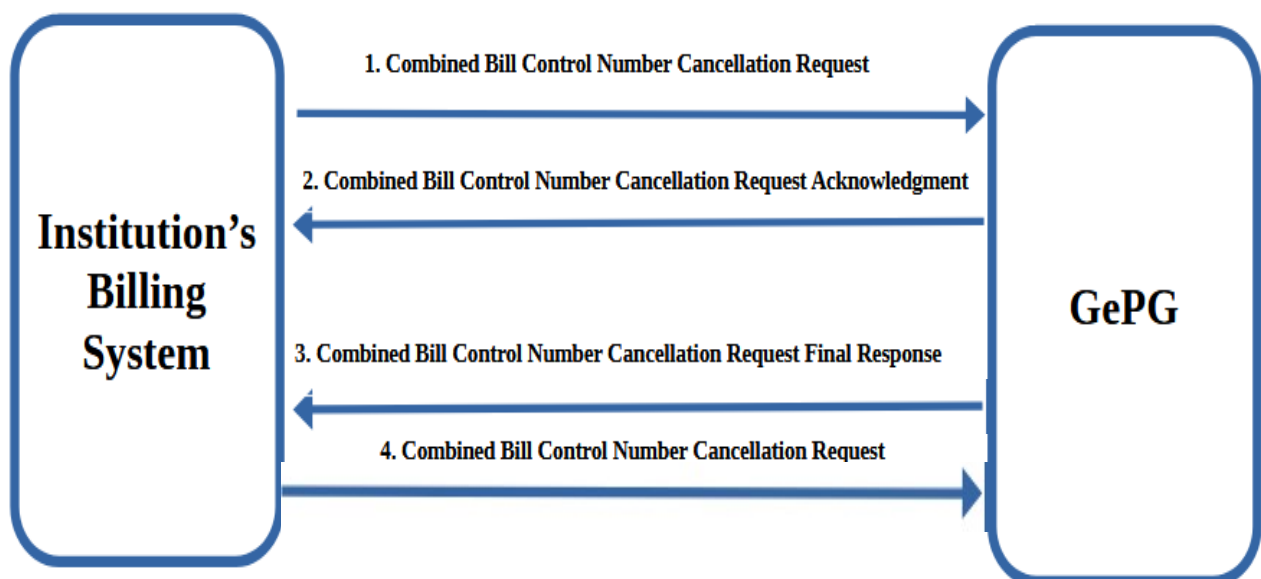
4.5 Bill Cancellation

Bill cancellation API enables Institutions to cancel pending bills when there is a situation that requires issued bills to be cancelled. This is a synchronous API where GePG will receive cancellation request, perform preliminary validations and cancel the identified bill. After cancellation GePG will return cancellation response to the requesting billing system within the same API call.

The cancelling Institution billing system will be required to submit unique bill identifier(**GrpBillId** for combined bill and **BillId** for normal bill) submitted earlier while requesting bill control number.

Upon cancellation of a combined bill control number, all individual bills forming the combined control number will be cancelled.

4.5.1 Combined Bill Control Number Cancellation Request

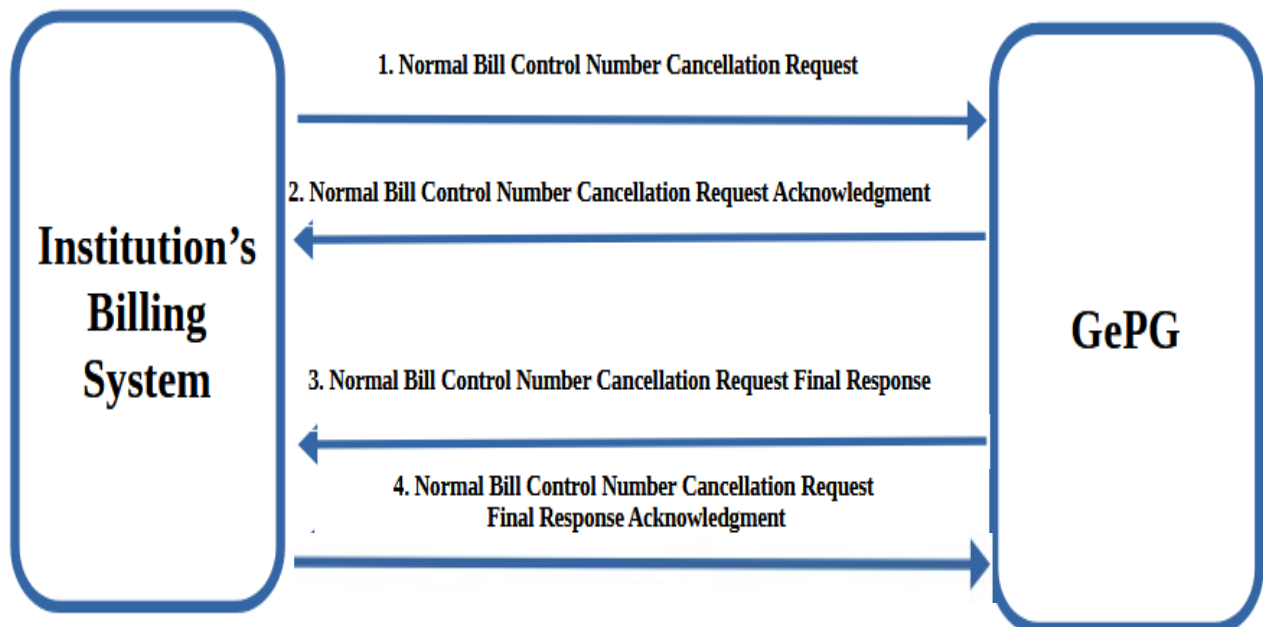


```

<Gepg>
  <billCanclReq>
    <ReqId>SP20210205130219</ReqId>
    <SpGrpCode>SPG1000</SpGrpCode>
    <SysCode>SYS001</SysCode>
    <BillTyp>2</BillTyp>
    <GrpBillId>PRI1581953689</GrpBillId>
    <CanclGenBy>Jamal Mwaika</CanclGenBy>
    <CanclApprBy>Hashim Abimeleck</CanclApprBy>
    <CanclReasn>Customer over billed</CanclReasn>
  </billCanclReq>
  <signature>SignatureGoesHere</signature>
</Gepg>

```

4.5.2 Normal Bill Control Number Cancellation Request



```

<Gepg>
  <billCanclReq>
    <ReqId>SP20210205130219</ReqId>
    <SpGrpCode>SP99128</SpGrpCode>
    <SysCode>SYS001</SysCode>
    <BillTyp>1</BillTyp>
    <GrpBillId>PRI1581953689</GrpBillId>
    <CanclGenBy>Jamal Mwaika</CanclGenBy>
    <CanclApprBy>Hashim Abimeleck</CanclApprBy>
    <CanclReasn>Customer over billed</CanclReasn>
  </billCanclReq>
  <signature>SignatureGoesHere</signature>
</Gepg>

```

4.5.3 Bill Cancellation Response

```
<Gepg>
  <billCanclRes>
    <ResId>GW20210205130219</ResId>
    <ReqId>SP20210205130219</ReqId>
    <GrpBillId>PRI1581953689</GrpBillId>
    <CanclSts>GS</CanclSts>
    <CanclStsCode>7283</CanclStsCode>
    <CanclStsDesc>Bill has been cancelled</CanclStsDesc>
  </billCanclRes>
  <signature>SignatureGoesHere</signature>
</Gepg>
```

4.6 Fields and parameters definition(s)

Tag	Data Type	Condition	Description
ReqId	String(100)	Mandatory	Service Provider Unique Bill Request Identification number
SpCode	String(10)	Mandatory	Service Provider Code
RtrRespFlg	Boolean	Mandatory	Return Response Flag - Boolean if True then response for the submit request is required.
BillId	String(100)	Mandatory	Unique identification of the Bill in Service Provider Billing System
SubSpCode	Numeric(10)	Mandatory	Service Provider Sub Code
CollCentCode	String(20)	Optional	Code identifying Service Provider Collection Centre
SysCode	String(10)	Mandatory	Service Provider System Identification Code
BillAmt	Numeric(32, 2)	Mandatory	Billed Amount can be decimal with precision of two.
MiscAmt	Numeric(32, 2)	Mandatory	Miscellaneous Amount can be decimal with precision of two. Always with billed currency. Default:0.00
BillExprDt	DateAndTime	Mandatory	Bill Expiry Date with format YYYY-MM-DDTHH24:MI:SS (2017-03-25T14:30:12).Beyond this date the bill is deemed invalid
CustId	String(50)	Mandatory	Customer Identification Reference
CustIdTyp	String	Mandatory	Customer Identification Reference Type
CustName	String(200)	Mandatory	Customer Name
BillDesc	String(500)	Optional	Bill Description
BillGenDt	DateAndTime	Mandatory	Bill Generation Date with format YYYY-MM-DDTHH24:MI:SS (2017-03-25T14:30:12) .

Tag	Data Type	Condition	Description
			Accepts today and past dates only.
BillGenBy	String (30)	Optional	Bill Generated By
BillApprBy	String (30)	Optional	Bill Approved By
CustCellNum	String (12)	Optional	Customer Mobile/Cell Number should have twelve digits including country code e.g. 255XXXXXXXXX
CustEmail	String (30)	Optional	Customer Email Address
Ccy	String (3)	Mandatory	Billed Amount Currency must be standard currency code
ExchRate	Numeric(32, 2)	Mandatory	Billed Amount Currency Exchange Rate Value to Tanzanian Shillings
BillEqvAmt	Numeric(32, 2)	Mandatory	Billed Equivalent Amount should be equal to the summation of all item equivalent amounts. Always submitted as TZS.
RemFlag	Boolean	Mandatory	Reminder Flag - True if reminder must be sent to the billed entity.
BillPayOpt	Numeric (1)	Mandatory	Bill Payment Options define the rules with which the bill should be paid. Can have the following values:- - o 1: FULL Bill- must be paid in one instalment with the amount paid equal or greater than the billed amount. - o 2: PARTIAL Bill- may be paid in multiple instalments with the last instalment greater or equal to the remaining billed amount. - o 3: EXACT Bill- must be paid

Tag	Data Type	Condition	Description
			in one instalment with the amount paid exactly equal to the billed amount. o 4: INFINITY payment option - shall be paid with any amount greater than zero (0). o 5: LIMITED payment option - may be paid in single or multiple instalments, with the last instalment equal to the remaining billed amount.
MinPayAmt	Numeric (32, 2)	Optional	Minimum Bill Payable Amount- This field carries the minimum amount payable value of a specific bill. This field makes the most in PARTIAL, LIMITED and INFINITY bill pay options.
PayPlan	Numeric (1)	Optional	Bill Payment Plan (Default:1) o 1: POST-PAID - Bill issued to customer after assessment on the service to be offered o 2: PRE-PAID - Customer defines and submits payment for desired service. Service provider may issue the service as per amount paid. This is for services that issue tokens for pre-paid meters
PayLimTyp	Numeric (1)	Mandatory / Optional	Bill Payment Limitation Type o 1: Normal Payment (No limitation) - This bill can be paid in any commercial bank or transfer to Central Bank o 2: All Commercial Bank Payment Limitation - Bill with this value can only be paid at commercial bank when its value does not exceed the maximum amount

Tag	Data Type	Condition	Description
			set in <PayLimAmt>30000</PayLimAmt> tag or else transfer to central bank will be required o 3: Central Bank Limitation - This bill can only paid by deposit or transfer to Central bank accounts o 4 : Central Bank and Specific Commercial Bank Limitation - Bill with this value can only be paid at only specific commercial bank listed in tag <CollPsp>PSP007,PSP008</CollPsp> when its value does not exceed the maximum amount set in PayLimAmt>30000</PayLimAmt> tag or else transfer to central bank will be required o 5 : Specific Commercial Bank Limitation - Bill with this value can only be paid at only specific commercial bank listed in tag <CollPsp>PSP020,PSP007</CollPsp>
PayLimAmt	Numeric (32, 2)	Mandatory /Optional	Bill Payment Limitation Amount - This field carries the maximum limitation value for a transaction
CollPsp	String (100)	Mandatory /Optional	Collection Payment Service Provider - This field carries a list of only PSP Codes allowed to settle a specific bill.
BillItemRef	String (50)	Mandatory	Item reference as deemed relevant by institution e.g. student registration number, customer account etc.

Tag	Data Type	Condition	Description
UseItemRefOnPay	String (1)	Mandatory	The value should be "N"
BillItemAmt	Numeric(32, 2)	Mandatory	Bill item amount can be decimal with precision of two. Always with billed currency.
BillItemEqvAmt	Numeric(32, 2)	Mandatory	Bill item equivalent amount should be equal to TZS equivalent of the bill item amount. Can be decimal with precision of two.
BillItemMiscAmt	Numeric(32, 2)	Mandatory	Bill item miscellaneous Amount can be decimal with precision of two. Always with bill item currency. Default:0.00
GfsCode	String (20)	Mandatory	GFS code is Government Standard for coding revenue sources. Should be specified for each bill item.
AckId	String(50)	Mandatory	GePG Unique Acknowledgement Identification number
AckStsCode	String(100)	Mandatory	Acknowledgement Status Code If several will be separated by semicolon e.g. (7242;7201). Refer Appendix I for codes definition.
AckStsDesc	String(500)	Mandatory	Aknowlegement Status Description If several will be separated by semicolon e.g. (Invalid Data format;Failure).
ResId	String (100)	Mandatory	Unique Bill Response identification Number.
ResSts	String (2)	Mandatory	Transaction Status o GF - GePG Failure o GS - GePG Success

Tag	Data Type	Condition	Description
CustCntrNum	Numeric (20)	Mandatory	Customer Control Number
BillApprBy	String (30)	Mandatory	Bill cancellation Approval User Details
CanclReasn	String (255)	Mandatory	Bill Cancellation reason
BillStsCode	String(100)	Mandatory	Bill Status Code
BillStsDesc	String(500)	Mandatory	Bill Status Description
PayTyp	Number (1)	Mandatory	Payment Type 1 - Pay All Bill Control Numbers At Once 2 - Pay any Bill Control Number
CanclGenBy	String(30)	Mandatory	Individual initiated bill cancelation from sp system
CanclApprBy	String(30)	Mandatory	Individual initiated bill approved bill cancelation from sp system
CanclSts	String(2)	Mandatory	Bill Cancelation General Status o GF - GePG Failure o GS - GePG Success
CanclStsCode	String(100)	Mandatory	Bill Cancelation Status Code
CanclStsDesc	String(500)	Mandatory	Bill Cancelation Status Description

5. QUOTE POSTING

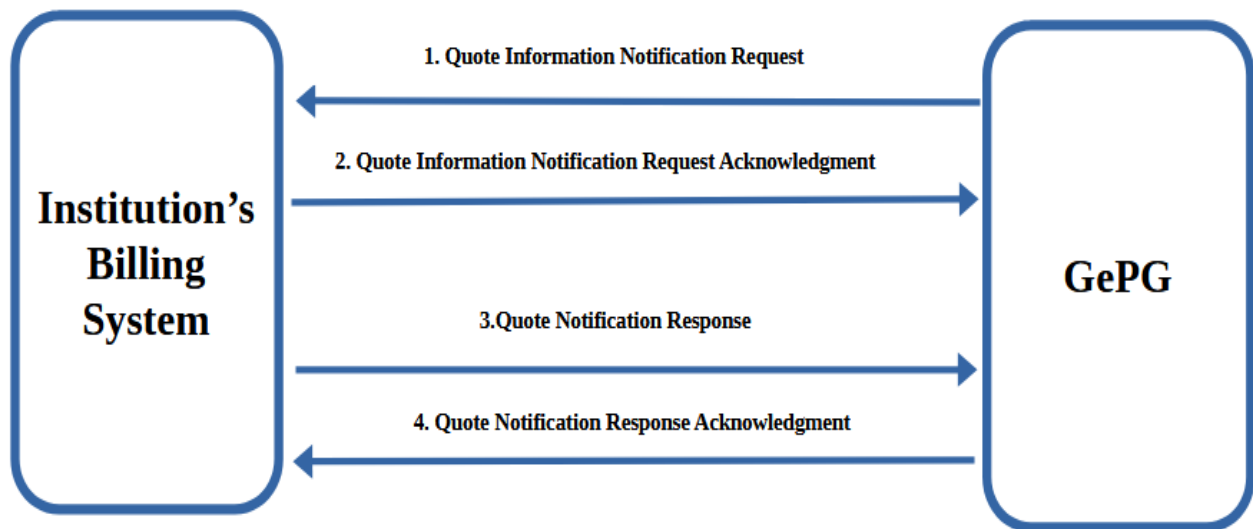
Quote posting refer to pre-paid services at which customer is issued with token to use service. Public institution bank account shall be credited upon successful generation of token.

5.1. Quote Information Posting Flow

- i. Once the quote is received, GePG will perform minimal validation. For valid request it will send quote information to a respective institution vending/billing system as per Quote Information Notification xml format, based on system indicated in *SysCode* field during bill submission.
- ii. Institution vending system receive the notification and immediately send back the acknowledgement before doing any internal processing to GePG as per Quote Information Notification Acknowledgment xml format
- iii. Institution vending system process the quote notification information received from GePG, compose the quote final response and send to GePG via provided call back url as per Quote Notification Response xml format.
- iv. The GePG system respond by acknowledge quote final response as per Quote Notification Response Acknowledgment xml format and.
- v. Once payment settle quote at GePG it will be sent to institution billing system as payment posting.

NOTE:

1. In case the GePG does not receive the acknowledgment or receive invalid xml format from vending system in step **ii.** above, it will keep on retrying to send the quote notification until the respective request aged 24 hours at which GePG will discard message.
2. In case vending system does not receive acknowledgement from GePG in step **iv.** above, it must keep retrying until acknowledgment is received. Unless there is a connectivity problem between the two system or any other problem.
3. In case billing system receive acknowledgement with response code other than **7101** in step **iv.** above, it should not perform retry until the returned code is acted accordingly. For example, response code **7242** implies **invalid xml format**, therefore retry can only be done after rectification of xml or data resulted invalid xml.



5.2. Quote Information Notification

The following xml for sending quote notification to vending/billing system from GePG

```

<Gepg>
  <quotSpNotfReq>
    <QuotHdr>
      <ReqId>d04221f6-dc24-402d-a811-ea7aec3263c4</ReqId>
      <SysCode>LXB001</SysCode>
      <SpCode>SP99108</SpCode>
      <CustCntrNum>0</CustCntrNum>
    </QuotHdr>
    <QuotDtl>
      <BillId>SP1158195368931</BillId>
      <BillCtrNum>991080175618</BillCtrNum>
      <PspCode>PSP006</PspCode>
      <PspName>Tanzani Commercial Bank</PspName>
      <QtRefId>254-283-85-234-5834-280</QtRefId>
      <RcptNum>990801756181111280</RcptNum>
      <PaidAmt>10.0</PaidAmt>
      <Ccy>TZS</Ccy>
      <CollAccNum>20101000015</CollAccNum>
      <TrxDtTm>2020-02-08T12:30:00</TrxDtTm>
      <UsdPayChnl>IB</UsdPayChnl>
      <TrdPtyTrxId>1039-39393320</TrdPtyTrxId>
      <PyrCellNum>255177100100</PyrCellNum>
      <PyrName>Luhaja Emmanuel</PyrName>
      <PyrEmail>e.luhaja@safeserva.cm</PyrEmail>
      <Rsv1/>
      <Rsv2/>
      <Rsv3/>
    </QuotDtl>
  </quotSpNotfReq>

```



```
<signature>SignatureGoesHere</signature>
</Gepg>
```

5.3. Quote Information Notification Acknowledgment

The following xml is for acknowledging quote notification, it returned by vending/billing system to GePG.

```
<Gepg>
  <quotSpNotfReqAck>
    <AckId>SP20210205130219</AckId>
    <ReqId>d04221f6-dc24-402d-a811-ea7aec3263c4</ReqId>
    <AckStsCode>7303</AckStsCode>
  </quotSpNotfReqAck>
  <signature>SignatureGoesHere</signature>
</Gepg>
```

5.4. Quote Notification Response

The following xml is for sending response of processed quotation to GePG from vending/billing system.

```
<Gepg>
  <quotSpRes>
    <QuotResHdr>
      <ReqId>202004241057562800167920114</ReqId>
      <SpCode>SP99108</SpCode>
      <SysCode>TPF001</SysCode>
      <CustCntrNum>991080175618</CustCntrNum>
      <QtStsCode>7101</QtStsCode>
      <QtStsDesc>Successful</QtStsDesc>
    </QuotResHdr>
    <QuotResDtl>
      <QtRefId>254-283-85-234-5834-280</QtRefId>
      <BillCtrNum>991080175618</BillCtrNum>
      <RcptNum>920108123456712</RcptNum>
      <PspCode>PSP006</PspCode>
      <PaidAmt>50000</PaidAmt>
      <TokenAmt>40983.61</TokenAmt>
      <Token>42352352354235325</Token>
      <Units>140.4KWH</Units>
      <DedItems>
        <DedItem>
          <DedCode>001</DedCode>
          <DedDesc>VAT</DedDesc>
          <DedAmt>7377.05</DedAmt>
          <DedPer>18%</DedPer>
        </DedItem>
        <DedItem>
          <DedCode>002</DedCode>
          <DedDesc>EWURA</DedDesc>
          <DedAmt>409.84</DedAmt>
          <DedPer>1%</DedPer>
        </DedItem>
      </DedItems>
    </QuotResDtl>
  </quotSpRes>
</Gepg>
```

```

        <DedItem>
            <DedCode>003</DedCode>
            <DedDesc>REA</DedDesc>
            <DedAmt>1229.50</DedAmt>
            <DedPer>3%</DedPer>
        </DedItem>
    </DedItems>
    <Rsv1/>
    <Rsv2/>
    <Rsv3/>
</QuotResDtl>
</quotSpRes>
<signature>SignatureGoesHere</signature>
</Gepg>

```

5.5. Quote Notification Response Acknowledgment

The following xml is for acknowledging quote response, it returned by GePG to vending/billing system.

```

<Gepg>
    <quotSpResAck>
        <AckId>SP20210205130219</AckId>
        <ReqId>202004241057562800167920114</ReqId>
        <AckStsCode>7303</AckStsCode>
        <AckStsDesc>Invalid message signature</AckStsDesc>
    </quotSpResAck>
    <signature>SignatureGoesHere</signature>
</Gepg>

```

Tag	Data Type	Condition	Description
ReqId	String (100)	Mandatory	Unique Identification of request
SysCode	String (10)	Mandatory	Service Provider System Identification Code
SpCode	String (10)	Mandatory	Service Provider Code
CustCntrNum	Number (20)	Mandatory	Customer Control Number
BillId	String (100)	Mandatory	Unique identification of the Bill in Service Provider Billing System
BillCntrNum	Number (20)	Mandatory	Bill Control Number
PspCode	String (10)	Mandatory	Payment Service Provider Code

Tag	Data Type	Condition	Description
PspName	String (300)	Mandatory	Payment Service Provider Name
QtRefId	String (100)	Mandatory	Unique Identification of quote transaction generated from PSP System
RcptNum	String (50)	Mandatory	Qualified GePG Receipt Number for the respective transaction.
PaidAmt	Numeric(32,2)	Mandatory	Paid Amount can be decimal with precision of two.
Ccy	String (3)	Mandatory	Currency
CollAccNum	String (50)	Mandatory	Collection Account Number to be Credited when quote is successful
TrxDtTm	DateAndTime	Mandatory	Transaction Date and Time YYYY-MM-DDTHH24:MI:SS (2021-04-17T14:30:12)
UsdPayChnl	String (30)	Mandatory	Used Payment Channel IB - Internet Banking CD - Cash Deposit TR - Account Transfer PS - Point of Sale MB - Mobile Banking OT - Online Transaction TIGOPESA- Tigo MPESA - Vodacom HALOPESA - Halotel TPESA - TTCL EAZYPESA - Zantel AIRTELMONEY-Airtel Money MC - Master Card VC - Visa Card Amex - American Express Card UPI - UnionPay
TrdPtyTrxId	String (255)	Optional	Third Party Receipt such as Issuing Bank authorization

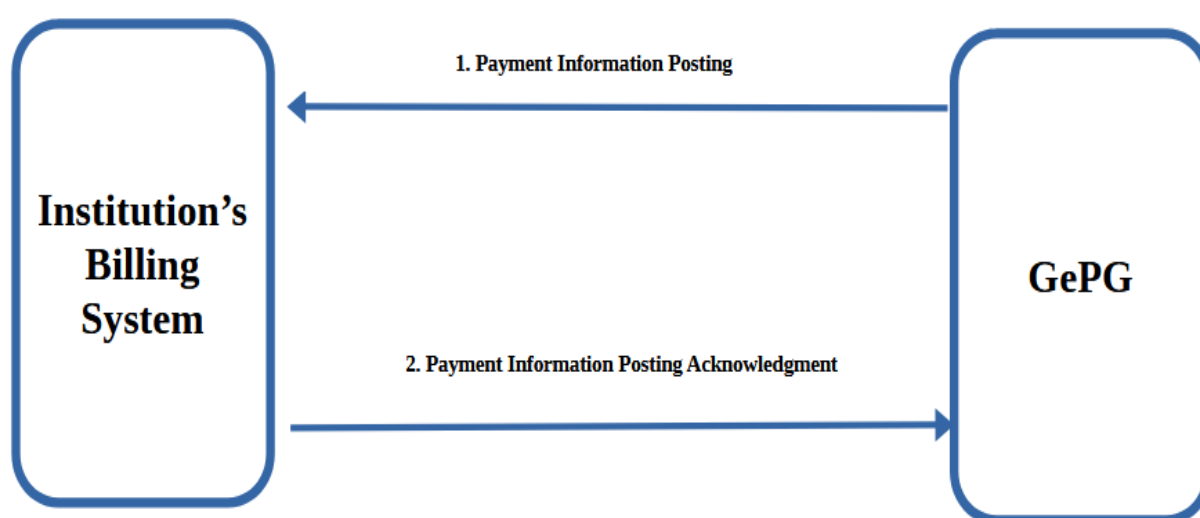
Tag	Data Type	Condition	Description
			Identification, MNO Receipt, Aggregator Receipt etc. This should be mandatory for all transaction done through third parties.
PyrCellNum	String (15)	Optional	Payer Mobile/Cell Number should start with a country code e.g. 255XXXXXXXXX
PyrName	String (1000)	Optional	Payer Name
PyrEmail	String (150)	Optional	Payer Email Address
Rsv1	String	Optional	Reserved field
Rsv2	String	Optional	Reserved field
Rsv3	String	Optional	Reserved field
AckId	String (100)	Mandatory	Unique Identification of acknowledgment
AckStsCode	String (100)	Mandatory	Acknowledgment Status Code
AckStsDesc	String (500)	Mandatory	Acknowledgment Status Code Description
QtStsCode	String (100)	Mandatory	Quote Status Code
QtStsDesc	String (500)	Mandatory	Quote Status description
TokenAmt	Numeric(32,2)	Optional	Amount of equivalent value with token generated
Token	Number	Optional	Utility Token
Units	Number	Optional	Number of Units Received
DedCode	String (50)	Optional	Deduction Code
DedDesc	String (100)	Optional	Deduction Description
DedAmt	Numeric(32,2)	Optional	Deduction Amount from Paid Amount
DedPer	String (50)	Optional	Percentage of Deduction

6. PAYMENT POSTING

Payment Service Provider (Bank, Mobile Money Operator) accepts the payment, credit the public institution bank account and share the payment information with GePG.

6.1. Payment Posting information Flow

- i. Once the payment is received, GePG will settle the bill and notify a respective institution by sending payment information as per Payment Information posting xml format to institution billing system indicated in *SysCode* field during bill submission.
- ii. The billing system must acknowledge to a received payment information with payload as per Payment Information posting Acknowledgement xml format before doing any internal processing.
- iii. NOTE:
 1. In case the GePG does not receive the acknowledgment or receive invalid xml format from institution billing system in step **iii.** above, it will keep on retrying to send the payment information until respective request aged 24 hours at which GePG will discard message.
 2. After the message has been discarded by GePG, the institution billing system will be required to use the reconciliation API to request payments of previous working day and settle respective bills.
 3. Reconciliation data last for seven (7) days from the date received from Bank at GePG.



6.2. Payment Information posting

The following xml is for sending payment information to institution billing system from GePG.

```
<Gepg>
  <pmtSpNtfReq>
    <PmtHdr>
      <ReqId>100000-2835934-85-23458343</ReqId>
      <GrpBillId>SP4099995368931</GrpBillId>
      <SpGrpCode>SPG40999</SpGrpCode>
      <CustCntrNum>409992345678</CustCntrNum>
      <EntryCnt>2</EntryCnt>
    </PmtHdr>
    <PmtDtls>
      <PmtTrxDtl>
        <SpCode>SP99842</SpCode>
        <BillId>10128879</BillId>
        <BillCtrNum>998420090050</BillCtrNum>
        <PspCode>PSP900</PspCode>
        <PspName>Tanzani Commercial Bank</PspName>
        <TrxId>100816320574</TrxId>
        <PayRefId>921033031098001</PayRefId>
        <BillAmt>12000</BillAmt>
        <PaidAmt>12000</PaidAmt>
        <BillPayOpt>3</BillPayOpt>
        <Ccy>TZS</Ccy>
        <CollAccNum>45610032210</CollAccNum>
        <TrxDtTm>2023-03-02T09:35:59</TrxDtTm>
        <UsdPayChnl>IB</UsdPayChnl>
        <TrdPtyTrxId>259-848-4898-88</TrdPtyTrxId>
        <QtRefId/>
        <PyrCellNum>255888662034</PyrCellNum>
        <PyrEmail/>
        <PyrName/>
        <Rsv1/>
        <Rsv2/>
        <Rsv3/>
      </PmtTrxDtl>
      <PmtTrxDtl>
        <SpCode>SP39842</SpCode>
        <BillId>1012887939</BillId>
        <BillCtrNum>398420090050</BillCtrNum>
        <PspCode>PSP900</PspCode>
        <PspName>Tanzani Commercial Bank</PspName>
        <TrxId>2100816320574</TrxId>
        <PayRefId>921033031098002</PayRefId>
        <BillAmt>20000</BillAmt>
        <PaidAmt>20000</PaidAmt>
        <BillPayOpt>3</BillPayOpt>
        <Ccy>TZS</Ccy>
        <CollAccNum>45610032209</CollAccNum>
        <TrxDtTm>2023-03-02T09:35:59</TrxDtTm>
        <UsdPayChnl>IB</UsdPayChnl>
        <TrdPtyTrxId>259-848-4898-88</TrdPtyTrxId>
```

```

        <QtRefId/>
        <PyrCellNum>255888662034</PyrCellNum>
        <PyrEmail/>
        <PyrName/>
        <Rsv1/>
        <Rsv2/>
        <Rsv3/>
    </PmtTrxDtl>
</PmtDtls>
</pmtSpNtfReq>
<signature>SignatureGoesHere</signature>
</Gepg>

```

6.3. Payment Information posting Acknowledgment

The following xml is for acknowledge payment information upon receive from GePG, it returned by institution billing system to GePG.

```

<Gepg>
  <pmtSpNtfReqAck>
    <AckId>SP20210205130219</AckId>
    <ReqId>100000-2835934-85-23458343</ReqId>
    <AckStsCode>7242</AckStsCode>
  </pmtSpNtfReqAck>
  <signature>SignatureGoesHere</signature>
</Gepg>

```

Tag	Data Type	Condition	Description
ReqId	String (100)	Mandatory	Unique Identification of notification generated by GePG
GrpBillId	String(100)	Mandatory	Unique identification of the Bill in Service Provider Billing System
SpGrpCode	String(10)	Mandatory	Service Provider Code for single or combined sps
EntryCnt	Number (4)	Mandatory	Total Number of payment entries contained in the details payload
SpCode	String (10)	Mandatory	Service Provider Code
BillId	String	Mandatory	Unique identification of the Bill

Tag	Data Type	Condition	Description
	(100)		in Service Provider Billing System
BillCtrNum	Number (12)	Mandatory	Paid Bill Control Number
PspCode	String (10)	Mandatory	Payment Service Provider Code
PspName	String (200)	Mandatory	Payment service provider Name eg. NMB, Vodacom etc
TrxId	String (100)	Mandatory	PSP Transaction Identification
PayRefId	String (100)	Mandatory	Payment receipt issued by the GePG
BillAmt	Numeric(3 2,2)	Mandatory	Bill Amount Billed Amount can be decimal with precision of two.
PaidAmt	Numeric(3 2,2)	Mandatory	Paid Amount can be decimal with precision of two.
BillPayOpt	String(10)	Mandatory	Bill Payment Options define the rules with which the bill should be paid. Can have the following values:- 1: FULL Bill- must be paid in one instalment with the amount paid equal or greater than the billed amount. 2: PARTIAL Bill- may be paid in multiple instalments with the last instalment greater or equal to the billed amount 3: EXACT Bill- must be paid in one instalment with the amount paid exactly equal to the billed amount. 4: INFINITY payment option - shall be paid with any amount greater than zero (0). 5: LIMITED payment option - may be paid in single or multiple

Tag	Data Type	Condition	Description
			instalments, with the last instalment equal to the remaining billed amount.
Ccy	String(3)	Mandatory	Paid Amount Currency
CollAccNum	String (50)	Mandatory	Credited Collection Account Number
TrxDtTm	DateAndTime	Mandatory	Transaction Date and Time with format YYYY-MM-DDTHH24:MI:SS (2017-03-25T14:30:12)
UsdPayChnl	String (50)	Mandatory	Payment provider payment channel used to pay the bill:- IB - Internet Banking CD - Cash Deposit TR - Account Transfer PS - Point of Sale MB - Mobile Banking OT - Online Transaction TIGOPESA- Tigo MPESA - Vodacom HALOPESA - Halotel TPESA - TTCL EAZYPESA - Zantel AIRTELMONEY-Airtel Money MC - Master Card VC - Visa Card Amex - American Express Card UPI - UnionPay
TrdPtyTrxId	String (255)	Optional	Third Party Receipt such as Issuing Bank authorization Identification, MNO Receipt, Aggregator Receipt etc.

Tag	Data Type	Condition	Description
QtRefId	String (100)	Optional	Unique Identification of quote transaction generated from PSP System
PyrCellNum	String (15)	Optional	Payer Mobile/Cell Number as received from payment service provider e.g. 255XXXXXXXXXX
PyrName	String (200)	Optional	Payer Name as received from payment service provider
PyrEmail	String (150)	Optional	Payer Email Address as received from payment service provider
Rsv1	String	Optional	Reserved field
Rsv2	String	Optional	Reserved field
Rsv3	String	Optional	Reserved field
AckId	String (100)	Mandatory	Unique Identification of acknowledgement
AckStsCode	String(100)	Mandatory	Acknowledgment Status Code

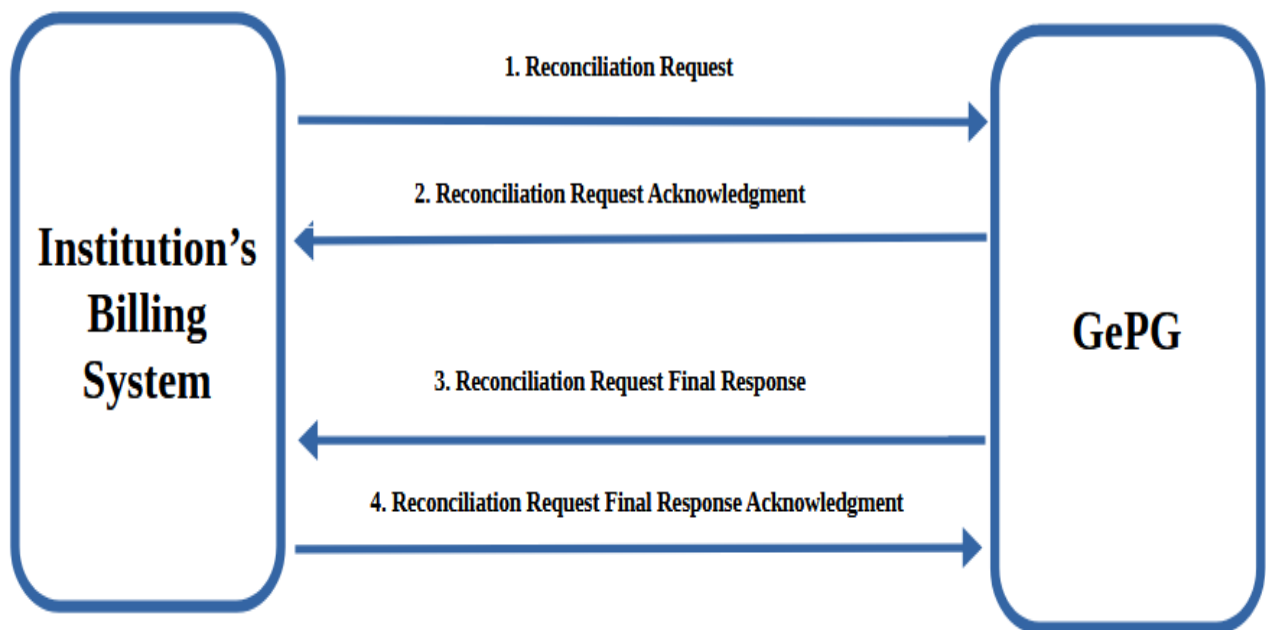
7. RECONCILIATION POSTING

7.1. Reconciliation Posting Information Flow

- i. Institution billing system sends reconciliation request to GePG by submitting payload as per Reconciliation Request xml format
- ii. GePG will receive the reconciliation request, perform minimal validations and acknowledge to the billing system as per Reconciliation Request Acknowledgement xml format.
- iii. GePG will then process the request by creating response payload as per Reconciliation Response and send it to the respective billing system.
- iv. GePG will receive the acknowledgement payload from billing system as per Reconciliation Response Acknowledgement.

NOTE:

1. Submission of reconciliation request from billing system to GePG of the previous day should start between 0600hrs to 2359hrs.
2. In case bill system does not receive acknowledgement from GePG in step **ii.** above, it must keep retrying until acknowledgment is received. Unless there is a connectivity problem between the two system or any other problem.
3. In case bill system receive acknowledgement with response code other than **7101** in step **ii.** above, it should not perform retry until the returned code is acted accordingly. For example, response code **7242** implies **invalid xml format**, therefore retry can only be done after rectification of xml or data resulted invalid xml.
4. Reconciliation data last for seven (7) days from the date received from Bank at GePG.



7.2. Reconciliation Request

The following xml is for sending reconciliation request to GePG from billing system.

```

Gepg>
  <sucSpPmtReq>
    <ReqId>d04221f6-dc24-402d-a811-ea7aec3263c4</ReqId>
    <SpGrpCode>SP99108</SpGrpCode>
    <SysCode>INST001</SysCode>
    <TrxDt>2023-03-09</TrxDt>
    <Rsv1/>
    <Rsv2/>
    <Rsv3/>
  </sucSpPmtReq>
  <signature>SignatureGoesHere</signature>
</Gepg>

```

7.3. Reconciliation Request Acknowledgment

The following xml is for acknowledging reconciliation request, it is returned by GePG to billing system.

```

<Gepg>
  <sucSpPmtReqAck>
    <AckId>SP20210205130219</AckId>
    <ReqId>d04221f6-dc24-402d-a811-ea7aec3263c4</ReqId>
    <AckStsCode>7303</AckStsCode>
    <AckStsDesc>Invalid message signature</AckStsDesc>
  </sucSpPmtReqAck>
</Gepg>

```

```

    </sucSpPmtReqAck>
    <signature>SignatureGoesHere</signature>
</Gepg>

```

7.4. Reconciliation Response

The following xml is for sending reconciliation response to institution billing system from GePG.

```

<Gepg>
  <sucSpPmtRes>
    <BatchHdr>
      <ResId>4747a838-67b7-11eb-ae93</ResId>
      <ReqId>d04221f6-dc24-402d-a811-ea7aec3263c4</ReqId>
      <SpGrpCode>SP99108</SpGrpCode>
      <SysCode>LXB001</SysCode>
      <PayStsCode>7101</PayStsCode>>
      <PayStsDesc>Sucess</PayStsDesc>
    </BatchHdr>
    <PmtDtls>
      <PmtTrxDtl>
        <CustCntrNum>994420090050</CustCntrNum>
        <GrpBillId>10128879</GrpBillId>
        <SpCode>SP99842</SpCode>
        <BillId>10128879</BillId>
        <BillCtrNum>998420090050</BillCtrNum>
        <PspCode>PSP900</PspCode>
        <PspName>Tanzani Commercial Bank</PspName>
        <TrxId>100816320574</TrxId>
        <PayRefId>921033031098001</PayRefId>
        <BillAmt>12000</BillAmt>
        <PaidAmt>12000</PaidAmt>
        <BillPayOpt>3</BillPayOpt>
        <Ccy>TZS</Ccy>
        <CollAccNum>45610032209</CollAccNum>
        <TrxDtTm>2023-03-02T09:35:59</TrxDtTm>
        <UsdPayChnl>IB</UsdPayChnl>
        <TrdPtyTrxId>259-848-4898-88</TrdPtyTrxId>
        <QtRefId/>
        <PyrCellNum>255888662034</PyrCellNum>
        <PyrEmail/>
        <PyrName/>
        <Rsv1/>
        <Rsv2/>
        <Rsv3/>
      </PmtTrxDtl>
      <PmtTrxDtl>
        <CustCntrNum>994420090050</CustCntrNum>
        <GrpBillId>10128879</GrpBillId>
        <SpCode>SP99842</SpCode>
        <BillId>10128879</BillId>
        <BillCtrNum>998420090050</BillCtrNum>
        <PspCode>PSP900</PspCode>
        <PspName>Tanzani Commercial Bank</PspName>
        <TrxId>100816320574</TrxId>
        <PayRefId>921033031098001</PayRefId>
        <BillAmt>12000</BillAmt>
        <PaidAmt>12000</PaidAmt>
      </PmtTrxDtl>
    </PmtDtls>
  </sucSpPmtRes>
</Gepg>

```

```

        <BillPayOpt>3</BillPayOpt>
        <Ccy>TZS</Ccy>
        <CollAccNum>45610032209</CollAccNum>
        <TrxDtTm>2023-03-02T09:35:59</TrxDtTm>
        <UsdPayChnl>IB</UsdPayChnl>
        <TrdPtyTrxId>259-848-4898-88</TrdPtyTrxId>
        <QtRefId/>
        <PyrCellNum>255888662034</PyrCellNum>
        <PyrEmail/>
        <PyrName/>
        <Rsv1/>
        <Rsv2/>
        <Rsv3/>
    </PmtTrxDtl>
</PmtDtls>
</sucSpPmtRes>
<signature>SignatureGoesHere</signature>
</Gepg>

```

7.5. Reconciliation Response Acknowledgment

The following xml is for acknowledging reconciliation response, it is returned by billing system to GePG.

```

<Gepg>
    <sucSpPmtResAck>
        <AckId>SP20230305130219</AckId>
        <ResId>4747a838-67b7-11eb-ae93</ResId>
        <AckStsCode>7101</AckStsCode>
    </sucSpPmtResAck>
    <signature>SignatureGoesHere</signature>
</Gepg>

```

Tag	Data Type	Condition	Description
ReqId	String (100)	Mandatory	Unique identification of request generaed by Service Provider System
SpGrpCode	String (10)	Mandatory	Service Provider Code or Combined Sps Code
SysCode	String (10)	Mandatory	Service Provider or Combined Sps System Identification Code
TrxDt	Date	Mandatory	Reconciliation transactions date with format YYYY-MM-DD (2023-03-25) Beyond this Tod the bill is deemed invalid

Tag	Data Type	Condition	Description
AckId	String (100)	Mandatory	Unique Identification of acknowledgement
AckStsCode	String(100)	Mandatory	Acknowledgment Status Code
AckStsDesc	String(500)	Mandatory	Acknowledgment Status Description
ResId	String (100)	Mandatory	Unique Response Identification generated by GePG
PayStsCode	String(100)	Mandatory	Reconciliation(Successful payment(s)) request Status Code
PayStsDesc	String(500)	Mandatory	Reconciliation(Successful payment(s)) request Status Description
GrpBillId	String (100)	Mandatory	Unique identification of the Bill in Service Provider Billing System
SpCode	String (10)	Mandatory	Service Provider Code
BillId	String (100)	Mandatory	Unique identification of the Bill in Service Provider Billing System
BillCtrNum	Number (12)	Mandatory	Paid Bill Control Number
PspCode	String (10)	Mandatory	Payment Service Provider Code
PspName	String (200)	Mandatory	Payment service provider Name eg. NMB, Vodacom etc
TrxId	String (100)	Mandatory	PSP Transaction Identification
PayRefId	String (100)	Mandatory	Payment receipt issued by the

Tag	Data Type	Condition	Description
			GePG
BillAmt	Numeric(32,2)	Mandatory	Bill Amount Billed Amount can be decimal with precision of two.
PaidAmt	Numeric(32,2)	Mandatory	Paid Amount can be decimal with precision of two.
BillPayOpt	String(10)	Mandatory	Bill Payment Options define the rules with which the bill should be paid. Can have the following values:- 8. 1: FULL Bill- must be paid in one instalment with the amount paid equal or greater than the billed amount. 9. 2: PARTIAL Bill- may be paid in multiple instalments with the last instalment greater or equal to the billed amount 10. 3: EXACT Bill- must be paid in one instalment with the amount paid exactly equal to the billed amount. 11. 4: INFINITY payment option - shall be paid with any amount greater than zero (0). 12. 5: LIMITED payment option - may be paid in single or multiple instalments, with the last instalment equal to the remaining billed amount.
Ccy	String(3)	Mandatory	Paid Amount Currency
CollAccNum	String (50)	Mandatory	Credited Collection Account Number
TrxDtTm	DateAndTime	Mandatory	Transaction Date and Time with format YYYY-MM-DDTHH24:MI:SS (2017-03-

Tag	Data Type	Condition	Description
			25T14:30:12)
UsdPayChnl	String (50)	Mandatory	Payment provider payment channel used to pay the bill:- IB - Internet Banking CD - Cash Deposit TR - Account Transfer PS - Point of Sale MB - Mobile Banking OT - Online Transaction TIGOPESA- Tigo MPESA - Vodacom HALOPESA - Halotel TPESA - TTCL EAZYPESA - Zantel AIRTELMONEY-Airtel Money MC - Master Card VC - Visa Card Amex - American Express Card UPI - UnionPay
TrdPtyTrxId	String (255)	Optional	Third Party Receipt such as Issuing Bank authorization Identification, MNO Receipt, Aggregator Receipt etc.
QtRefId	String (100)	Optional	Unique Identification of quote transaction generated from PSP System
PyrCellNum	String (15)	Optional	Payer Mobile/Cell Number as received from payment service provider e.g. 255XXXXXXXXX
PyrName	String (200)	Optional	Payer Name as received from payment service provider
PyrEmail	String (150)	Optional	Payer Email Address as received from payment service provider

Tag	Data Type	Condition	Description
Rsv1	String	Optional	Reserved field
Rsv2	String	Optional	Reserved field
Rsv3	String	Optional	Reserved field

13.DIGITAL SIGNATURE SECURITY

13.1. Overview

Digital signature will be used for messages exchange between the GePG and Service Provider. The SP will be required to apply for digital signature from Certificate Authority (CA). Digital signatures will also be used for message(s) sharing. Message(s) must previously be signed by the Participant's private key and verified using participant's Public Key.

13.2. Digital Signature

The security architecture used by the GePG is based on Public Key Infrastructure (PKI) technology in which two keys (Public and Private) are used to provide the highest levels of security and universal acceptance.

There are typically three algorithms involved with the digital signature process:

- a) Key generation – This algorithm provides a private key along with its corresponding public key from CA.
- b) Signing – This algorithm produces a signature upon receiving a private key and the message that is being signed.
- c) Verification – This algorithm checks for the authenticity of the message by verifying it along with the signature and the public key.

13.3. Digital Signature File Format

GePG accepts and shares public and private keys in Public-Key Cryptography Standards version 12 (PKCS #12). The expected filename extension for PKCS #12 files is .p12 or .pfx. For example, **spABCPublicKey.pfx** or **spABCPublicKey.p12**

13.4. Signature Algorithm

The supported signature algorithm is SHA256withRSA

13.5. Signed Final Payload Format

GePG expects message to be digitally signed and obey the following rules:

Sample content to be signed

```
<billCanclReq>
  <ReqId>SP20210205130219</ReqId>
  <SpGrpCode>SP99128</SpGrpCode>
  <SysCode>SYS001</SysCode>
  <BillTyp>1</BillTyp>
  <GrpBillId>PRI1581953689</GrpBillId>
  <CanclGenBy>Jamal Mwaika</CanclGenBy>
  <CanclApprBy>Hashim Abimeleck</CanclApprBy>
  <CanclReasn>Customer over billed</CanclReasn>
</billCanclReq>
```

Final payload after signing

```
<Gepg>
  <billCanclReq>
    <ReqId>SP20210205130219</ReqId>
    <SpGrpCode>SP99128</SpGrpCode>
    <SysCode>SYS001</SysCode>
    <BillTyp>1</BillTyp>
    <GrpBillId>PRI1581953689</GrpBillId>
    <CanclGenBy>Jamal Mwaika</CanclGenBy>
    <CanclApprBy>Hashim Abimeleck</CanclApprBy>
    <CanclReasn>Customer over billed</CanclReasn>
  </billCanclReq>
  <signature>U6vJ6jZYnaMq0Jf5itoNOv1l/+m+0JfNO1l=</signature>
</Gepg>
```

14. APPENDIX

Content under this section may be updated from time to time and still be used on the same API version. SP may request for an updated version if available.

14.1. Customer Identity Types

The following shall be used by SP as Customer Identity Type captured (Identity value) submitted on the respective payload

Identity Value	Identity Description
1	National Identification Number
2	Driver's License
3	Tax Payer's Identification
4	Wallet Pay Number

14.2. Response Codes

List of response codes maybe added but prior information will be provided.

CODE	DESCRIPTION
7101	Successful
7201	Failure
7202	Required header is not given
7203	Unauthorized
7204	Bill does not exist
7205	Invalid service provider
7206	Service provider is not active
7207	Duplicate payment
7208	Invalid business account
7209	Business account is not active
7210	Collection account balance limit reached
7211	Payment service provider code did not match Bill service provider code

CODE	DESCRIPTION
7212	Payment currency did not match Bill currency
7213	Bill has expired
7214	Insufficient amount paid
7215	Invalid payment service provider
7216	Payment service provider is not active
7219	Wrong currency
7241	Invalid session
7242	Invalid request data
7279	Bill already settled
7281	Invalid payment transaction date
7284	Payment currency did not match collection account currency
7289	Invalid batch generated date
7290	Total batch amount cannot be zero(0)
7291	Total batch amount is not equal to summation of items(transactions)
7297	Inactive currency
7301	Bill has been paid partially
7302	Paid amount is not exact billed amount
7303	Invalid Signature
7304	Invalid Signature Configuration missing one of parameter
7305	Invalid batch start and end date
7306	Batch has no item(transaction)
7307	Inconsistency batch start, end and generated date
7311	Invalid GePG configurations missing one of parameter

CODE	DESCRIPTION
7317	Error on processing request
7318	Invalid reconciliation request date
7319	Reconciliation request date is out of allowable range
7321	Request cannot complete at this time, try later
7322	Inactive communication protocol
7323	Invalid code, mismatch of supplied code on information and header
7333	Invalid bill control number
7335	Requested bill control number does not exist
7336	Requested bill amount has reached the maximum payable limit
7337	Requested resource is not available